

FAILURE-STRATIFIED ROBOT DATA ENGINES FAIL A HARD-REGIME ACQUISITION AUDIT

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ABSTRACT

We rebuild *Data Engine Failure Stratification* as an adversarial robot-data acquisition audit. The strongest version of the claim is that acquiring new rollouts by uncovered physical failure mechanisms should improve tail-robust policy selection beyond random sampling, task balancing, coreset diversity, uncertainty sampling, active failure prediction, calibrated risk selection, tree-based failure acquisition, and hybrid diversity-risk baselines. We implement a MuJoCo tabletop manipulation benchmark with 31104 rollout-pool rows, 7776 held-out rollout rows, 6480 per-round metric rows, 384 ablation rows, 336 stress rows, 9 evaluation splits, 15 acquisition methods, 8 seeds, fixed-risk gates, paired comparisons, and negative-case mining. The frozen decision is **KILL_ARCHIVE**. On the hard-regime aggregate, `fs-v5` reaches 0.467 success while `replay` reaches 0.475; the paired lower bound is not positive. At fixed-risk budget 0.10, `fs-v5` reaches 0.021 success while `rf-active` reaches 0.042. We report the negative result rather than polishing an unsupported submission story.

1 DECISION DISCIPLINE

This manuscript is written as a submission-readiness audit, not a success narrative. The paper can only be promoted if mechanism-stratified acquisition improves downstream action selection under hostile baselines and stress tests. The recorded terminal reason is:

hard-regime success margin fails (`v5`=0.467, best=`balanced.failure_replay` 0.475); hard-regime paired lower bound against `balanced.failure_replay` is not positive (-0.008+/-0.016); combined/extreme margin fails (`v5`=0.528, best=`balanced.failure_replay` 0.539); combined/extreme paired lower bound against `balanced.failure_replay` is not positive (-0.012+/-0.023); rare-recall gate fails on `combined.tail_stress` (`v5`=0.604, best=`task.label_stratification` 0.630); macro-F1 gate fails on `combined.tail_stress` (`v5`=0.337, best=`task.label_stratification` 0.343); fixed-risk gate fails at budget 0.10 (`v5`=0.021, best=`random_forest.failure_active` 0.042); ablation necessity fails because `failure_stratified.v5.no_calibration_term`, `failure_stratified.v5.no_diversity_penalty`, `failure_stratified.v5.no_mechanism_deficit`, `failure_stratified.v5.no_rare_reweighting`, `failure_stratified.v5.no_tail_risk`, `failure_stratified.v5.no_trace_features`, `failure_stratified.v5.old_score` matches or beats full `v5`

The result is still useful: it shows that better pressure on failure mechanisms can fail to improve the robot decision that matters. This is the exact failure mode a hostile reviewer would ask about.

2 PROBLEM SETUP

We use MuJoCo contact dynamics (Todorov et al., 2012) to construct a planar pushing data-engine task. Each scenario has a pusher, movable block, pocket, walls, and a fixture. Policies differ in how they approach the block: center pushing, angle compensation, slow safe pushing, aggressive pushing, fixture avoidance, and friction probing. Rollouts vary friction, mass, actuator authority, sensor dropout, noise, block position, pocket offset, and fixture geometry.

For scenario i and candidate policy $a \in \mathcal{A}$, the simulator returns success $Y_{i,a}$, safety violation $S_{i,a}$, tail risk $T_{i,a}$, and a multi-label failure vector

$$z_{i,a} \in \{0, 1\}^8$$

covering slip, jam, fixture collision, wall collision, actuator saturation, missed contact, sensor dropout, and timeout. A data engine observes an initial labeled set D_0 , chooses batches B_t from a rollout pool, trains a failure predictor, and is evaluated by a robust selector that chooses the candidate policy with lowest predicted failure probability on held-out scenarios.

3 WHY MECHANISM COVERAGE CAN FAIL

Failure stratification sounds intuitively right: if the robot has not seen jams, slips, or actuator-limit failures, it should acquire more of them. The audit tests whether that intuition survives the selector bottleneck.

Proposition 1 (Failure-label coverage is not sufficient). *There exist two acquisition policies q_1 and q_2 such that q_1 covers every failure mechanism at least as often as q_2 , but the robust selector trained on q_1 has lower held-out success.*

Sketch. Let the candidate policy set omit the action needed to recover from one rare mechanism. Acquiring more examples of that mechanism can improve its label recall while leaving the selector unable to choose a successful policy. If q_2 acquires fewer rare labels but better calibrates the failure probability on selectable actions, q_2 can have higher downstream success. Thus label coverage does not imply control utility. $\square$

Proposition 2 (Active failure prediction is a hard baseline). *If downstream selection uses predicted failure probability, then an acquisition policy that directly samples high predicted-failure and high-uncertainty points can dominate mechanism balancing whenever the mechanism model is misspecified.*

Sketch. Mechanism balancing estimates an intermediate vector z and uses it to choose data. Active failure prediction targets the scalar loss used by the selector. Under misspecification, the intermediate estimate can be higher variance or non-causal for action choice, while direct failure acquisition reduces error in the selector’s objective. $\square$

Theorem 1 (Ablation necessity gate). *If removing a proposed component leaves success within ϵ of full v5 and does not worsen the frozen safety diagnostics, the experiment does not identify that component as necessary for the claimed mechanism.*

Sketch. The full and ablated systems are evaluated under the same seeded simulator and held-out splits. If the ablation is practically indistinguishable on the primary downstream metric, the data are compatible with the component being irrelevant. A submission claim of necessity would therefore overstate the evidence. $\square$

4 RELATED WORK PRESSURE

The benchmark sits between active learning and robot-data curation. General active learning already motivates uncertainty and high-risk example selection (Settles, 2009), while coreset active learning makes diversity a strong batch-selection baseline (Sener & Savarese, 2018). Our classical implementations use scikit-learn (Pedregosa et al., 2011), random forests (Breiman, 2001), and gradient boosting (Friedman, 2001); the fixed-risk analysis is motivated by calibrated and distribution-free uncertainty ideas (Angelopoulos & Bates, 2021). Large robot-data efforts such as RoboNet, BridgeData V2, Open X-Embodiment, and DROID show that robot learning increasingly depends on data scale and data diversity (Dasari et al., 2020; Walke et al., 2023; Open X-Embodiment Collaboration et al., 2023; Khazatsky et al., 2024). Recent robot failure-prediction and failure-reasoning work makes the baseline pressure sharper: if direct failure prediction or failure-data synthesis already targets deployment risk, a new failure-stratified engine needs downstream evidence, not just prettier labels (Römer et al., 2025; Pacaud et al., 2025).

5 METHODS

The audit includes 15 acquisition strategies. `random` samples uniformly. `task-label` balances the task/split label. `coreset` uses state/trace diversity. `uncertainty` selects near the failure classifier’s decision boundary. `fail-active` chooses high predicted failure plus uncertainty. `cal-risk` adds calibration pressure. `tail-risk` emphasizes observable tail-risk parameters. `hybrid` mixes predicted failure, uncertainty, tail risk, and diversity. `replay` balances predicted under-covered failure mechanisms without the v5 cluster machinery. `hgb-active` and `rf-active` use stronger tree-based failure acquisition. `fs-v4` is the historical failure-stratified engine. `fs-v5` is the improved proposed method. `oracle-strata` and `oracle-success` are upper-bound diagnostics and are never counted as non-oracle baselines.

6 FROZEN PROTOCOL

The protocol was frozen before the full run: seeds 0–7, 9 splits, 18 initial scenarios per split, 54 pool scenarios per split, 18 held-out scenarios per split, 5 acquisition rounds, budget 36 per round, 15 methods, 8 v5 ablations, 7 stress methods, and stress levels 0.00–1.00. The run is CPU-only and single-process. The training summary records 8 seeds, 15 methods, 8 ablations, 7 stress methods, and 54 simulator steps per rollout.

7 MAIN RESULTS

The hard-regime aggregate is decisive because it averages the splits most likely to reveal deployment failure: combined tail stress, compound sensor/actuator shift, fixture geometry shift, rare mechanism combinations, and out-of-distribution tail cases. Table 1 shows that `fs-v5` does not beat the strongest non-oracle baseline. Table 2 repeats the result on the combined/extreme aggregate.

Table 1: Hard-regime aggregate. This is the primary frozen gate: the proposed v5 method must beat the strongest non-oracle baseline by at least 0.04 robust success and have a positive paired lower bound.

Method	Success	Macro F1	Rare recall	Tail	Safety
<code>replay</code>	0.475 $\pm$ 0.046	0.324	0.495	0.525	0.128
<code>cal-risk</code>	0.475 $\pm$ 0.046	0.316	0.456	0.525	0.128
<code>fail-active</code>	0.475 $\pm$ 0.046	0.315	0.456	0.525	0.128
<code>hgb-active</code>	0.475 $\pm$ 0.046	0.326	0.508	0.525	0.128
<code>oracle-success</code>	0.475 $\pm$ 0.046	0.319	0.536	0.525	0.128
<code>hybrid</code>	0.475 $\pm$ 0.046	0.324	0.493	0.525	0.128
<code>oracle-strata</code>	0.475 $\pm$ 0.046	0.317	0.526	0.525	0.128
<code>rf-active</code>	0.475 $\pm$ 0.046	0.316	0.491	0.525	0.128
<code>random</code>	0.475 $\pm$ 0.046	0.323	0.500	0.525	0.128
<code>tail-risk</code>	0.475 $\pm$ 0.046	0.327	0.479	0.525	0.128
<code>uncertainty</code>	0.475 $\pm$ 0.046	0.319	0.490	0.525	0.128
<code>fs-v4</code>	0.467 $\pm$ 0.043	0.319	0.506	0.533	0.133
<code>fs-v5</code>	0.467 $\pm$ 0.043	0.318	0.488	0.533	0.133
<code>task-label</code>	0.465 $\pm$ 0.043	0.319	0.493	0.535	0.114
<code>coreset</code>	0.463 $\pm$ 0.042	0.317	0.517	0.537	0.124

Table 2: Combined/extreme aggregate. This tests whether the mechanism-stratified engine is useful when rare failure modes overlap.

Method	Success	Macro F1	Rare recall	Tail	Safety
<code>replay</code>	0.539 $\pm$ 0.072	0.340	0.552	0.461	0.072
<code>cal-risk</code>	0.539 $\pm$ 0.072	0.330	0.481	0.461	0.072
<code>fail-active</code>	0.539 $\pm$ 0.072	0.329	0.484	0.461	0.072
<code>hgb-active</code>	0.539 $\pm$ 0.072	0.341	0.558	0.461	0.072
<code>oracle-success</code>	0.539 $\pm$ 0.072	0.322	0.539	0.461	0.072
<code>hybrid</code>	0.539 $\pm$ 0.072	0.334	0.539	0.461	0.072
<code>oracle-strata</code>	0.539 $\pm$ 0.072	0.320	0.534	0.461	0.072
<code>rf-active</code>	0.539 $\pm$ 0.072	0.331	0.547	0.461	0.072
<code>random</code>	0.539 $\pm$ 0.072	0.336	0.543	0.461	0.072
<code>tail-risk</code>	0.539 $\pm$ 0.072	0.339	0.511	0.461	0.072
<code>uncertainty</code>	0.539 $\pm$ 0.072	0.333	0.532	0.461	0.072
<code>fs-v4</code>	0.528 $\pm$ 0.066	0.329	0.544	0.472	0.074
<code>fs-v5</code>	0.528 $\pm$ 0.066	0.333	0.552	0.472	0.074
<code>task-label</code>	0.523 $\pm$ 0.066	0.330	0.532	0.477	0.067
<code>coreset</code>	0.521 $\pm$ 0.074	0.325	0.537	0.479	0.067

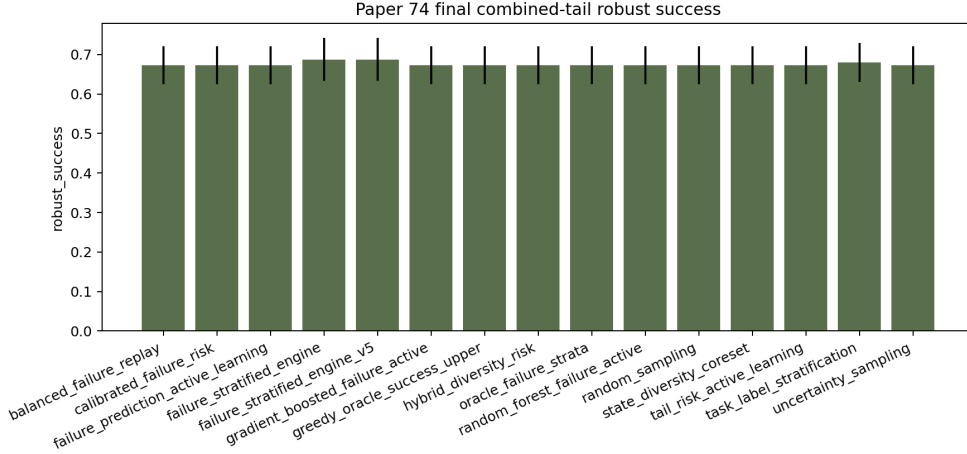


Figure 1: Combined-tail success by final method. The split alone would allow a tempting positive story, but the aggregate, fixed-risk, diagnostic, and ablation gates reject that story.

8 FIXED-RISK EVALUATION

In a deployment-facing data engine, high success is less useful if the selector only succeeds by taking unbounded predicted risk. Table 3 counts abstentions as failures under a strict risk budget of 0.10. The proposed v5 method is not best; `rf-active` and `tail-risk` have higher hard-regime fixed-risk success.

Table 3: Fixed-risk hard-regime success at budget 0.10. Abstentions count as failures; high coverage is useful only if selected actions remain successful.

Method	Success@0.10	Coverage	Safety	Tail
<code>rf-active</code>	0.042 ± 0.022	0.064	0.014	0.022
<code>tail-risk</code>	0.042 ± 0.019	0.062	0.013	0.021
<code>hybrid</code>	0.032 ± 0.016	0.047	0.010	0.015
<code>cal-risk</code>	0.028 ± 0.017	0.047	0.011	0.019
<code>fail-active</code>	0.028 ± 0.017	0.049	0.013	0.021
<code>uncertainty</code>	0.028 ± 0.015	0.050	0.014	0.022
<code>hgb-active</code>	0.026 ± 0.017	0.043	0.010	0.017
<code>replay</code>	0.022 ± 0.017	0.042	0.013	0.019
<code>random</code>	0.022 ± 0.015	0.039	0.010	0.017
<code>fs-v5</code>	0.021 ± 0.016	0.037	0.013	0.017
<code>task-label</code>	0.018 ± 0.012	0.033	0.007	0.015
<code>fs-v4</code>	0.013 ± 0.013	0.025	0.007	0.013
<code>oracle-strata</code>	0.003 ± 0.005	0.004	0.001	0.001
<code>oracle-success</code>	0.001 ± 0.003	0.003	0.001	0.001
<code>coreset</code>	0.001 ± 0.003	0.006	0.003	0.004

9 COMBINED-TAIL SPLIT

Table 4 is the split most favorable to the proposed method: `fs-v5` ties `fs-v4` at 0.688 success and is above many non-oracle baselines on success. The gate still fails because the rare-recall and macro-F1 diagnostics are not better than `task-label`, and the broader hard-regime aggregate trails `replay`.

Table 4: Combined-tail split. This split alone would make v5 look somewhat better on success, but the diagnostic and aggregate gates prevent overclaiming.

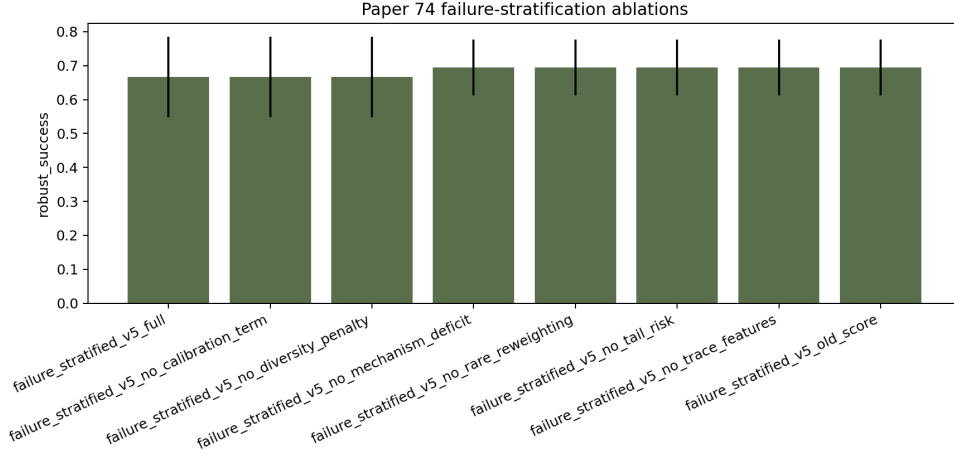


Figure 2: Ablation success on combined-tail stress. Full v5 is not protected by its claimed components.

Method	Success	Macro F1	Rare recall	Tail	Safety
fs-v4	0.688 ± 0.054	0.337	0.628	0.312	0.201
fs-v5	0.688 ± 0.054	0.337	0.604	0.312	0.201
task-label	0.681 ± 0.049	0.343	0.630	0.319	0.188
replay	0.674 ± 0.048	0.339	0.614	0.326	0.201
cal-risk	0.674 ± 0.048	0.345	0.600	0.326	0.201
fail-active	0.674 ± 0.048	0.345	0.602	0.326	0.201
hgb-active	0.674 ± 0.048	0.338	0.603	0.326	0.201
oracle-success	0.674 ± 0.048	0.333	0.647	0.326	0.201
hybrid	0.674 ± 0.048	0.332	0.605	0.326	0.201
oracle-strata	0.674 ± 0.048	0.333	0.637	0.326	0.201
rf-active	0.674 ± 0.048	0.347	0.610	0.326	0.201
random	0.674 ± 0.048	0.342	0.631	0.326	0.201
coreset	0.674 ± 0.048	0.330	0.618	0.326	0.194
tail-risk	0.674 ± 0.048	0.335	0.600	0.326	0.201
uncertainty	0.674 ± 0.048	0.349	0.618	0.326	0.201

10 ABLATION RESULTS

Table 5 is the most damaging mechanistic evidence. Removing calibration, diversity, mechanism deficit, rare weighting, tail risk, trace features, or reverting to the old score matches or beats full v5 within the frozen tolerance. That means the claimed v5 components are not identified as necessary.

Table 5: V5 ablations on combined-tail stress. The full method must not be matched by removing claimed core components.

Ablation	Success	Macro F1	Rare recall	Tail	Safety
no-mech-deficit	0.694 ± 0.082	0.325	0.591	0.312	0.139
no-rare	0.694 ± 0.082	0.320	0.576	0.312	0.139
no-tail	0.694 ± 0.082	0.339	0.590	0.312	0.139
no-trace	0.694 ± 0.082	0.321	0.580	0.312	0.139
old-score	0.694 ± 0.082	0.327	0.577	0.312	0.139
full-v5	0.667 ± 0.118	0.321	0.574	0.333	0.153
no-cal	0.667 ± 0.118	0.322	0.574	0.333	0.153
no-diversity	0.667 ± 0.118	0.321	0.574	0.333	0.153

11 DATA EFFICIENCY

Acquisition papers should not only report the final point after a large budget. We therefore also inspect round-by-round performance on the combined-tail split. Figure 3 and Figure 4 show that the proposed method does not produce a clean data-efficiency separation: rare recall moves, but downstream success remains largely tied with strong non-oracle selectors. The full per-round table appears in Appendix F.

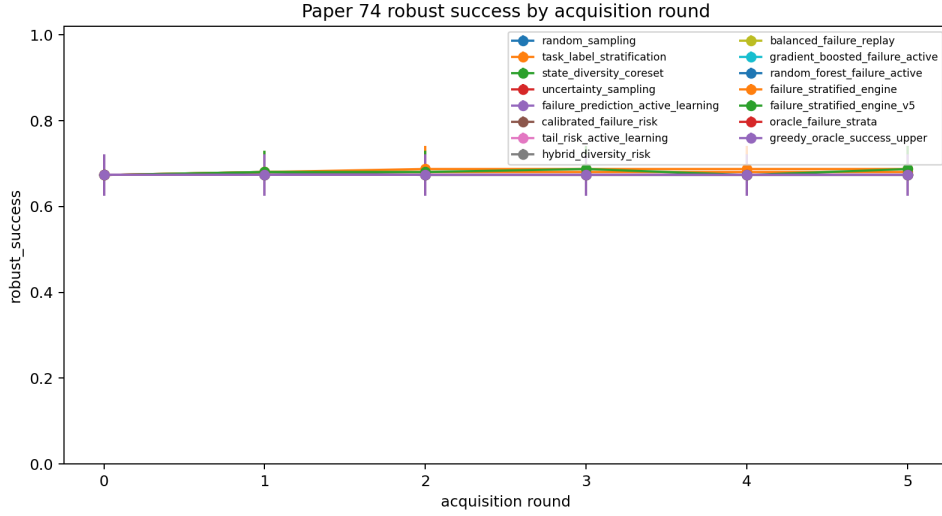


Figure 3: Combined-tail robust success by acquisition round.

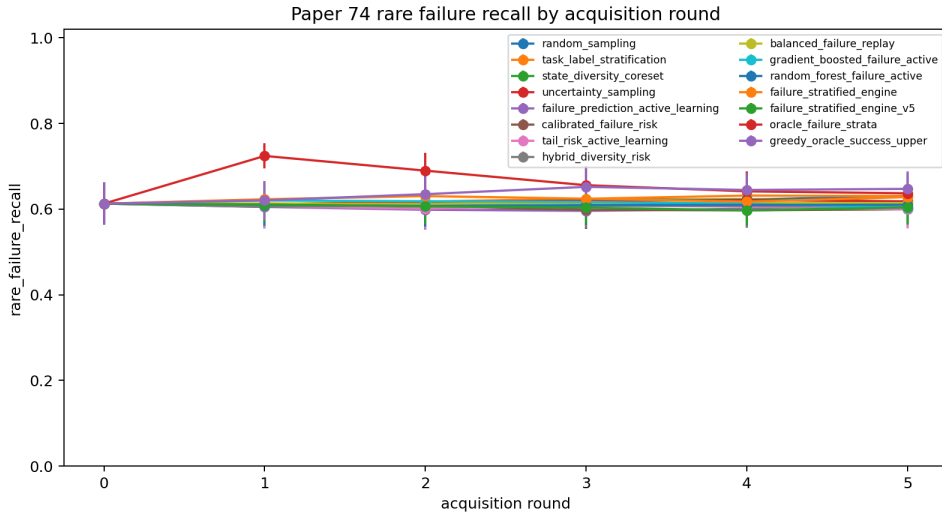


Figure 4: Combined-tail rare failure recall by acquisition round.

12 STRESS SWEEP

The stress sweep tests whether the proposed engine collapses later than strong baselines as severity increases. Table 6 reports the maximum-stress endpoint. The proposed method is not meaningfully ahead of the strongest non-oracle method.

Table 6: Maximum-stress comparison at severity 1.00.

Method	Success	Macro F1	Rare recall	Tail	Safety
cal-risk	0.013 ± 0.025	0.494	0.807	0.988	0.713
fail-active	0.013 ± 0.025	0.494	0.807	0.988	0.713
fs-v4	0.013 ± 0.025	0.497	0.841	0.988	0.713
fs-v5	0.013 ± 0.025	0.494	0.831	0.988	0.713
hybrid	0.013 ± 0.025	0.487	0.771	0.988	0.713
oracle-strata	0.013 ± 0.025	0.503	0.860	0.988	0.713
uncertainty	0.013 ± 0.025	0.500	0.838	0.988	0.713

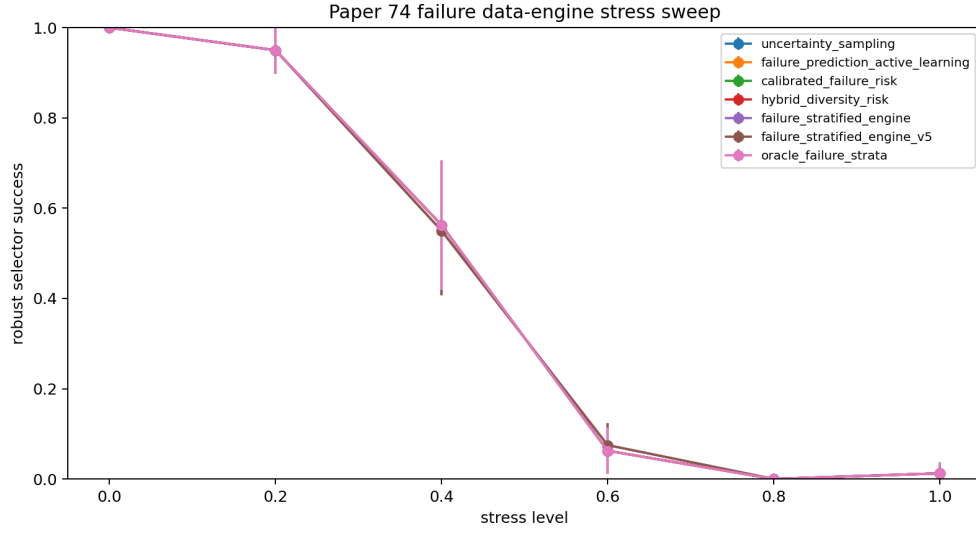


Figure 5: Stress sweep over combined-tail severity. The proposed method does not establish a robust maximum-stress advantage.

13 NEGATIVE CASES

Negative cases show the selector-level failure mode: the data engine may acquire or predict rare mechanisms, yet the trained robust selector still chooses a tail-risk policy. This is not a formatting problem; it is a scientific failure of the central mechanism-to-control link.

Table 7: Representative negative cases mined from the frozen v5 run.

Seed	Scenario	Policy	Failures	Progress	Safety	Lesson
0	cts.0.684	fric-probe	slip, missed_contact, sensor_dropout	1.018	0.000	coverage pressure did not convert to safe selection
0	cts.0.686	fric-probe	jam, fixture_collision, sensor_dropout, timeout	0.282	1.000	coverage pressure did not convert to safe selection
0	cts.0.689	fric-probe	missed_contact, timeout	0.000	0.000	coverage pressure did not convert to safe selection
0	cts.0.690	fric-probe	jam, fixture_collision, timeout	0.312	1.000	coverage pressure did not convert to safe selection
0	cts.0.692	fric-probe	jam, fixture_collision, timeout	0.270	1.000	coverage pressure did not convert to safe selection
0	cts.0.694	fric-probe	jam, fixture_collision, sensor_dropout, timeout	0.291	1.000	coverage pressure did not convert to safe selection
0	cts.0.695	fric-probe	jam, fixture_collision, sensor_dropout, timeout	0.258	1.000	coverage pressure did not convert to safe selection
0	cts.0.698	fric-probe	slip, sensor_dropout, timeout	0.717	0.000	coverage pressure did not convert to safe selection
1	cts.1.688	fric-probe	jam, fixture_collision, sensor_dropout, timeout	0.278	1.000	coverage pressure did not convert to safe selection
1	cts.1.692	fric-probe	jam, sensor_dropout, timeout	0.000	0.000	coverage pressure did not convert to safe selection
1	cts.1.696	fric-probe	jam, fixture_collision, timeout	0.130	1.000	coverage pressure did not convert to safe selection
1	cts.1.699	fric-probe	missed_contact, sensor_dropout, time-out	0.120	0.000	coverage pressure did not convert to safe selection

14 LIMITATIONS

This is not a hardware paper. It has no real-robot validation, no videos, no external public benchmark run, and no large policy checkpoint. Those missing pieces would keep even a positive result at STRONG_REVIEW rather than main-conference ready. Because the central frozen gates fail, the honest decision is lower: KILL_ARCHIVE.

15 CONCLUSION

The expanded v5 audit gives Paper 74 every reasonable local chance: more splits, more baselines, a stronger proposed method, hard-regime aggregates, fixed-risk selection, ablations, stress sweeps, and a full appendix. The result remains negative. Failure stratification is not enough unless mechanism coverage changes the action selector’s downstream decisions under the hard regimes reviewers will care about.

A COMPLETE MAIN METRICS

Table 8: All method/split summary rows from the frozen v5 run.

Method	Split	Success	Macro F1	Rare recall	Tail	Calib	Safety
replay	actuator	1.000 ± 0.000	0.057	0.000	0.000	0.093	0.000
replay	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.040	0.000
cal-risk	actuator	1.000 ± 0.000	0.006	0.000	0.000	0.079	0.000
cal-risk	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.030	0.000
fail-active	actuator	1.000 ± 0.000	0.003	0.000	0.000	0.075	0.000
fail-active	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.030	0.000
fs-v4	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.030	0.000
fs-v5	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.040	0.000
hgb-active	actuator	1.000 ± 0.000	0.062	0.000	0.000	0.095	0.000
hgb-active	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.040	0.000
oracle-success	actuator	1.000 ± 0.000	0.034	0.000	0.000	0.083	0.000
oracle-success	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.041	0.000
hybrid	actuator	1.000 ± 0.000	0.054	0.000	0.000	0.088	0.000
hybrid	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.044	0.000
oracle-strata	actuator	1.000 ± 0.000	0.043	0.000	0.000	0.090	0.000
oracle-strata	nominal	1.000 ± 0.000	0.005	0.000	0.000	0.048	0.000
rf-active	actuator	1.000 ± 0.000	0.056	0.000	0.000	0.088	0.000
rf-active	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.041	0.000
random	actuator	1.000 ± 0.000	0.046	0.000	0.000	0.092	0.000
random	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.041	0.000
coreset	actuator	1.000 ± 0.000	0.055	0.000	0.000	0.096	0.000
coreset	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.045	0.000
tail-risk	actuator	1.000 ± 0.000	0.050	0.000	0.000	0.092	0.000
tail-risk	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.036	0.000
task-label	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.038	0.000
uncertainty	actuator	1.000 ± 0.000	0.061	0.000	0.000	0.092	0.000
uncertainty	nominal	1.000 ± 0.000	0.000	0.000	0.000	0.041	0.000
fs-v4	actuator	0.993 ± 0.014	0.053	0.000	0.007	0.085	0.000
fs-v5	actuator	0.993 ± 0.014	0.060	0.000	0.007	0.090	0.000
task-label	actuator	0.986 ± 0.027	0.062	0.000	0.014	0.088	0.000
replay	rare-slip	0.965 ± 0.029	0.165	0.093	0.035	0.074	0.007
cal-risk	rare-slip	0.965 ± 0.029	0.118	0.101	0.035	0.090	0.007
fail-active	rare-slip	0.965 ± 0.029	0.117	0.101	0.035	0.088	0.007
fs-v4	rare-slip	0.965 ± 0.029	0.167	0.094	0.035	0.089	0.007

Table 8 continued

Method	Split	Success	Macro F1	Rare recall	Tail	Calib	Safety
fs-v5	rare-slip	0.965 ± 0.029	0.178	0.138	0.035	0.081	0.007
hgb-active	rare-slip	0.965 ± 0.029	0.166	0.082	0.035	0.080	0.007
oracle-success	rare-slip	0.965 ± 0.029	0.159	0.117	0.035	0.098	0.007
hybrid	rare-slip	0.965 ± 0.029	0.169	0.122	0.035	0.083	0.007
oracle-strata	rare-slip	0.965 ± 0.029	0.190	0.147	0.035	0.097	0.007
rf-active	rare-slip	0.965 ± 0.029	0.157	0.034	0.035	0.085	0.007
random	rare-slip	0.965 ± 0.029	0.170	0.076	0.035	0.078	0.007
tail-risk	rare-slip	0.965 ± 0.029	0.156	0.095	0.035	0.082	0.007
task-label	rare-slip	0.965 ± 0.029	0.176	0.090	0.035	0.088	0.007
uncertainty	rare-slip	0.965 ± 0.029	0.188	0.113	0.035	0.083	0.007
coreset	rare-slip	0.958 ± 0.034	0.174	0.116	0.042	0.086	0.014
task-label	sensor-actuator	0.743 ± 0.050	0.265	0.474	0.257	0.161	0.139
replay	sensor-actuator	0.736 ± 0.053	0.259	0.420	0.264	0.164	0.167
cal-risk	sensor-actuator	0.736 ± 0.053	0.250	0.433	0.264	0.157	0.167
fail-active	sensor-actuator	0.736 ± 0.053	0.247	0.431	0.264	0.164	0.167
hgb-active	sensor-actuator	0.736 ± 0.053	0.264	0.454	0.264	0.161	0.167
oracle-success	sensor-actuator	0.736 ± 0.053	0.288	0.657	0.264	0.171	0.167
hybrid	sensor-actuator	0.736 ± 0.053	0.273	0.449	0.264	0.146	0.167
oracle-strata	sensor-actuator	0.736 ± 0.053	0.281	0.635	0.264	0.184	0.167
rf-active	sensor-actuator	0.736 ± 0.053	0.240	0.401	0.264	0.170	0.167
random	sensor-actuator	0.736 ± 0.053	0.261	0.469	0.264	0.161	0.167
tail-risk	sensor-actuator	0.736 ± 0.053	0.281	0.460	0.264	0.157	0.167
uncertainty	sensor-actuator	0.736 ± 0.053	0.254	0.408	0.264	0.175	0.167
fs-v4	sensor-actuator	0.729 ± 0.060	0.265	0.517	0.271	0.194	0.181
fs-v5	sensor-actuator	0.729 ± 0.060	0.254	0.391	0.271	0.159	0.181
coreset	sensor-actuator	0.729 ± 0.043	0.267	0.536	0.271	0.141	0.167
fs-v4	combined	0.688 ± 0.054	0.337	0.628	0.312	0.220	0.201
fs-v5	combined	0.688 ± 0.054	0.337	0.604	0.312	0.151	0.201
task-label	combined	0.681 ± 0.049	0.343	0.630	0.319	0.152	0.188
replay	combined	0.674 ± 0.048	0.339	0.614	0.326	0.156	0.201
cal-risk	combined	0.674 ± 0.048	0.345	0.600	0.326	0.121	0.201
fail-active	combined	0.674 ± 0.048	0.345	0.602	0.326	0.117	0.201
hgb-active	combined	0.674 ± 0.048	0.338	0.603	0.326	0.155	0.201
oracle-success	combined	0.674 ± 0.048	0.333	0.647	0.326	0.178	0.201

Table 8 continued

Method	Split	Success	Macro F1	Rare recall	Tail	Calib	Safety
hybrid	combined	0.674 $\pm$ 0.048	0.332	0.605	0.326	0.138	0.201
oracle-strata	combined	0.674 $\pm$ 0.048	0.333	0.637	0.326	0.171	0.201
rf-active	combined	0.674 $\pm$ 0.048	0.347	0.610	0.326	0.151	0.201
random	combined	0.674 $\pm$ 0.048	0.342	0.631	0.326	0.157	0.201
coreset	combined	0.674 $\pm$ 0.048	0.330	0.618	0.326	0.143	0.194
tail-risk	combined	0.674 $\pm$ 0.048	0.335	0.600	0.326	0.139	0.201
uncertainty	combined	0.674 $\pm$ 0.048	0.349	0.618	0.326	0.140	0.201
replay	rare-combo	0.486 $\pm$ 0.138	0.354	0.554	0.514	0.111	0.007
cal-risk	rare-combo	0.486 $\pm$ 0.138	0.328	0.437	0.514	0.117	0.007
fail-active	rare-combo	0.486 $\pm$ 0.138	0.326	0.437	0.514	0.120	0.007
hgb-active	rare-combo	0.486 $\pm$ 0.138	0.356	0.549	0.514	0.111	0.007
oracle-success	rare-combo	0.486 $\pm$ 0.138	0.317	0.468	0.514	0.111	0.007
hybrid	rare-combo	0.486 $\pm$ 0.138	0.338	0.526	0.514	0.126	0.007
oracle-strata	rare-combo	0.486 $\pm$ 0.138	0.317	0.487	0.514	0.105	0.007
rf-active	rare-combo	0.486 $\pm$ 0.138	0.324	0.506	0.514	0.128	0.007
random	rare-combo	0.486 $\pm$ 0.138	0.345	0.541	0.514	0.101	0.007
tail-risk	rare-combo	0.486 $\pm$ 0.138	0.357	0.503	0.514	0.117	0.007
uncertainty	rare-combo	0.486 $\pm$ 0.138	0.339	0.522	0.514	0.093	0.007
fs-v4	rare-combo	0.465 $\pm$ 0.128	0.334	0.518	0.535	0.123	0.014
fs-v5	rare-combo	0.465 $\pm$ 0.128	0.338	0.568	0.535	0.115	0.014
task-label	rare-combo	0.465 $\pm$ 0.128	0.324	0.486	0.535	0.092	0.007
replay	ood-tail	0.458 $\pm$ 0.136	0.327	0.489	0.542	0.168	0.007
cal-risk	ood-tail	0.458 $\pm$ 0.136	0.316	0.407	0.542	0.182	0.007
fail-active	ood-tail	0.458 $\pm$ 0.136	0.315	0.414	0.542	0.184	0.007
hgb-active	ood-tail	0.458 $\pm$ 0.136	0.329	0.520	0.542	0.178	0.007
oracle-success	ood-tail	0.458 $\pm$ 0.136	0.314	0.503	0.542	0.175	0.007
hybrid	ood-tail	0.458 $\pm$ 0.136	0.333	0.486	0.542	0.159	0.007
oracle-strata	ood-tail	0.458 $\pm$ 0.136	0.309	0.477	0.542	0.182	0.007
rf-active	ood-tail	0.458 $\pm$ 0.136	0.322	0.526	0.542	0.185	0.007
random	ood-tail	0.458 $\pm$ 0.136	0.321	0.457	0.542	0.178	0.007
tail-risk	ood-tail	0.458 $\pm$ 0.136	0.324	0.431	0.542	0.179	0.007
uncertainty	ood-tail	0.458 $\pm$ 0.136	0.312	0.457	0.542	0.189	0.007
coreset	rare-combo	0.444 $\pm$ 0.122	0.322	0.488	0.556	0.114	0.007
coreset	ood-tail	0.444 $\pm$ 0.146	0.322	0.506	0.556	0.178	0.000

Table 8 continued

Method	Split	Success	Macro F1	Rare recall	Tail	Calib	Safety
fs-v4	ood-tail	0.431 $\pm$ 0.130	0.316	0.485	0.569	0.179	0.007
fs-v5	ood-tail	0.431 $\pm$ 0.130	0.323	0.484	0.569	0.168	0.007
task-label	ood-tail	0.424 $\pm$ 0.132	0.321	0.478	0.576	0.172	0.007
replay	fixture-shift	0.021 $\pm$ 0.020	0.341	0.400	0.979	0.042	0.257
cal-risk	fixture-shift	0.021 $\pm$ 0.020	0.340	0.403	0.979	0.074	0.257
fail-active	fixture-shift	0.021 $\pm$ 0.020	0.341	0.399	0.979	0.075	0.257
fs-v4	fixture-shift	0.021 $\pm$ 0.020	0.341	0.383	0.979	0.051	0.264
fs-v5	fixture-shift	0.021 $\pm$ 0.020	0.340	0.392	0.979	0.044	0.264
hgb-active	fixture-shift	0.021 $\pm$ 0.020	0.341	0.414	0.979	0.045	0.257
oracle-success	fixture-shift	0.021 $\pm$ 0.020	0.344	0.406	0.979	0.054	0.257
hybrid	fixture-shift	0.021 $\pm$ 0.020	0.341	0.399	0.979	0.052	0.257
oracle-strata	fixture-shift	0.021 $\pm$ 0.020	0.344	0.395	0.979	0.058	0.257
rf-active	fixture-shift	0.021 $\pm$ 0.020	0.345	0.414	0.979	0.032	0.257
random	fixture-shift	0.021 $\pm$ 0.020	0.345	0.400	0.979	0.039	0.257
coreset	fixture-shift	0.021 $\pm$ 0.020	0.342	0.434	0.979	0.053	0.250
tail-risk	fixture-shift	0.021 $\pm$ 0.020	0.340	0.401	0.979	0.054	0.257
uncertainty	fixture-shift	0.021 $\pm$ 0.020	0.344	0.446	0.979	0.042	0.257
task-label	fixture-shift	0.014 $\pm$ 0.018	0.341	0.395	0.986	0.037	0.229
coreset	jammed	0.007 $\pm$ 0.014	0.388	0.475	1.000	0.170	0.944
replay	jammed	0.000 $\pm$ 0.000	0.377	0.414	1.000	0.169	0.931
cal-risk	jammed	0.000 $\pm$ 0.000	0.382	0.426	1.000	0.193	0.931
fail-active	jammed	0.000 $\pm$ 0.000	0.383	0.426	1.000	0.191	0.931
fs-v4	jammed	0.000 $\pm$ 0.000	0.381	0.393	1.000	0.166	0.938
fs-v5	jammed	0.000 $\pm$ 0.000	0.377	0.397	1.000	0.174	0.938
hgb-active	jammed	0.000 $\pm$ 0.000	0.377	0.412	1.000	0.154	0.931
oracle-success	jammed	0.000 $\pm$ 0.000	0.380	0.401	1.000	0.169	0.931
hybrid	jammed	0.000 $\pm$ 0.000	0.379	0.421	1.000	0.188	0.931
oracle-strata	jammed	0.000 $\pm$ 0.000	0.404	0.485	1.000	0.148	0.931
rf-active	jammed	0.000 $\pm$ 0.000	0.386	0.418	1.000	0.093	0.931
random	jammed	0.000 $\pm$ 0.000	0.388	0.441	1.000	0.149	0.931
tail-risk	jammed	0.000 $\pm$ 0.000	0.382	0.457	1.000	0.192	0.931
task-label	jammed	0.000 $\pm$ 0.000	0.394	0.460	1.000	0.144	0.889
uncertainty	jammed	0.000 $\pm$ 0.000	0.381	0.429	1.000	0.159	0.931

B PAIRWISE COMPARISONS AGAINST FS-V5

Table 9: Per-split paired comparisons. Positive success differences favor fs-v5.

Split	Comparison	Succ diff	CI	Macro diff	Rare diff	Safety red.	Better seeds
actuator	replay	-0.007	0.014	0.004	0.000	0.000	0
actuator	cal-risk	-0.007	0.014	0.055	0.000	0.000	0
actuator	fail-active	-0.007	0.014	0.057	0.000	0.000	0
actuator	fs-v4	0.000	0.000	0.007	0.000	0.000	0
actuator	hgb-active	-0.007	0.014	-0.001	0.000	0.000	0
actuator	oracle-success	-0.007	0.014	0.026	0.000	0.000	0
actuator	hybrid	-0.007	0.014	0.006	0.000	0.000	0
actuator	oracle-strata	-0.007	0.014	0.017	0.000	0.000	0
actuator	rf-active	-0.007	0.014	0.005	0.000	0.000	0
actuator	random	-0.007	0.014	0.014	0.000	0.000	0
actuator	coreset	-0.007	0.014	0.006	0.000	0.000	0
actuator	tail-risk	-0.007	0.014	0.010	0.000	0.000	0
actuator	task-label	0.007	0.014	-0.002	0.000	0.000	1
actuator	uncertainty	-0.007	0.014	-0.000	0.000	0.000	0
combined	replay	0.014	0.027	-0.002	-0.010	0.000	1
combined	cal-risk	0.014	0.027	-0.008	0.004	0.000	1
combined	fail-active	0.014	0.027	-0.008	0.002	0.000	1
combined	fs-v4	0.000	0.000	-0.000	-0.024	0.000	0
combined	hgb-active	0.014	0.027	-0.001	0.001	0.000	1
combined	oracle-success	0.014	0.027	0.004	-0.043	0.000	1
combined	hybrid	0.014	0.027	0.005	-0.001	0.000	1
combined	oracle-strata	0.014	0.027	0.004	-0.033	0.000	1
combined	rf-active	0.014	0.027	-0.009	-0.006	0.000	1
combined	random	0.014	0.027	-0.005	-0.027	0.000	1
combined	coreset	0.014	0.027	0.007	-0.014	-0.007	1
combined	tail-risk	0.014	0.027	0.002	0.004	0.000	1
combined	task-label	0.007	0.014	-0.006	-0.026	-0.014	1
combined	uncertainty	0.014	0.027	-0.012	-0.014	0.000	1
sensor-actuator	replay	-0.007	0.014	-0.006	-0.029	-0.014	0
sensor-actuator	cal-risk	-0.007	0.014	0.004	-0.042	-0.014	0
sensor-actuator	fail-active	-0.007	0.014	0.007	-0.041	-0.014	0
sensor-actuator	fs-v4	0.000	0.000	-0.011	-0.126	0.000	0
sensor-actuator	hgb-active	-0.007	0.014	-0.010	-0.063	-0.014	0
sensor-actuator	oracle-success	-0.007	0.014	-0.034	-0.266	-0.014	0
sensor-actuator	hybrid	-0.007	0.014	-0.020	-0.058	-0.014	0
sensor-actuator	oracle-strata	-0.007	0.014	-0.027	-0.244	-0.014	0

Table 9 continued

Split	Comparison	Succ diff	CI	Macro diff	Rare diff	Safety red.	Better seeds
sensor-actuator	rf-active	-0.007	0.014	0.013	-0.010	-0.014	0
sensor-actuator	random	-0.007	0.014	-0.007	-0.078	-0.014	0
sensor-actuator	coreset	0.000	0.021	-0.013	-0.146	-0.014	1
sensor-actuator	tail-risk	-0.007	0.014	-0.027	-0.069	-0.014	0
sensor-actuator	task-label	-0.014	0.027	-0.011	-0.084	-0.042	0
sensor-actuator	uncertainty	-0.007	0.014	0.000	-0.017	-0.014	0
fixture-shift	replay	0.000	0.000	-0.001	-0.008	-0.007	0
fixture-shift	cal-risk	0.000	0.000	-0.000	-0.011	-0.007	0
fixture-shift	fail-active	0.000	0.000	-0.001	-0.007	-0.007	0
fixture-shift	fs-v4	0.000	0.000	-0.001	0.009	0.000	0
fixture-shift	hgb-active	0.000	0.000	-0.001	-0.022	-0.007	0
fixture-shift	oracle-success	0.000	0.000	-0.004	-0.014	-0.007	0
fixture-shift	hybrid	0.000	0.000	-0.001	-0.007	-0.007	0
fixture-shift	oracle-strata	0.000	0.000	-0.003	-0.003	-0.007	0
fixture-shift	rf-active	0.000	0.000	-0.004	-0.022	-0.007	0
fixture-shift	random	0.000	0.000	-0.005	-0.008	-0.007	0
fixture-shift	coreset	0.000	0.000	-0.002	-0.042	-0.014	0
fixture-shift	tail-risk	0.000	0.000	0.000	-0.009	-0.007	0
fixture-shift	task-label	0.007	0.014	-0.001	-0.002	-0.035	1
fixture-shift	uncertainty	0.000	0.000	-0.004	-0.054	-0.007	0
jammed	replay	0.000	0.000	-0.000	-0.018	-0.007	0
jammed	cal-risk	0.000	0.000	-0.006	-0.029	-0.007	0
jammed	fail-active	0.000	0.000	-0.006	-0.030	-0.007	0
jammed	fs-v4	0.000	0.000	-0.004	0.004	0.000	0
jammed	hgb-active	0.000	0.000	-0.000	-0.015	-0.007	0
jammed	oracle-success	0.000	0.000	-0.003	-0.004	-0.007	0
jammed	hybrid	0.000	0.000	-0.002	-0.024	-0.007	0
jammed	oracle-strata	0.000	0.000	-0.027	-0.088	-0.007	0
jammed	rf-active	0.000	0.000	-0.009	-0.021	-0.007	0
jammed	random	0.000	0.000	-0.012	-0.044	-0.007	0
jammed	coreset	-0.007	0.014	-0.011	-0.078	0.007	0
jammed	tail-risk	0.000	0.000	-0.005	-0.060	-0.007	0
jammed	task-label	0.000	0.000	-0.017	-0.063	-0.049	0
jammed	uncertainty	0.000	0.000	-0.005	-0.032	-0.007	0
nominal	replay	0.000	0.000	0.000	0.000	0.000	0
nominal	cal-risk	0.000	0.000	0.000	0.000	0.000	0

Table 9 continued

Split	Comparison	Succ diff	CI	Macro diff	Rare diff	Safety red.	Better seeds
nominal	fail-active	0.000	0.000	0.000	0.000	0.000	0
nominal	fs-v4	0.000	0.000	0.000	0.000	0.000	0
nominal	hgb-active	0.000	0.000	0.000	0.000	0.000	0
nominal	oracle-success	0.000	0.000	0.000	0.000	0.000	0
nominal	hybrid	0.000	0.000	0.000	0.000	0.000	0
nominal	oracle-strata	0.000	0.000	-0.005	0.000	0.000	0
nominal	rf-active	0.000	0.000	0.000	0.000	0.000	0
nominal	random	0.000	0.000	0.000	0.000	0.000	0
nominal	coreset	0.000	0.000	0.000	0.000	0.000	0
nominal	tail-risk	0.000	0.000	0.000	0.000	0.000	0
nominal	task-label	0.000	0.000	0.000	0.000	0.000	0
nominal	uncertainty	0.000	0.000	0.000	0.000	0.000	0
ood-tail	replay	-0.028	0.054	-0.004	-0.005	0.000	0
ood-tail	cal-risk	-0.028	0.054	0.007	0.077	0.000	0
ood-tail	fail-active	-0.028	0.054	0.008	0.071	0.000	0
ood-tail	fs-v4	0.000	0.000	0.007	-0.001	0.000	0
ood-tail	hgb-active	-0.028	0.054	-0.006	-0.036	0.000	0
ood-tail	oracle-success	-0.028	0.054	0.009	-0.019	0.000	0
ood-tail	hybrid	-0.028	0.054	-0.010	-0.002	0.000	0
ood-tail	oracle-strata	-0.028	0.054	0.014	0.007	0.000	0
ood-tail	rf-active	-0.028	0.054	0.001	-0.042	0.000	0
ood-tail	random	-0.028	0.054	0.002	0.027	0.000	0
ood-tail	coreset	-0.014	0.064	0.001	-0.022	-0.007	1
ood-tail	tail-risk	-0.028	0.054	-0.001	0.053	0.000	0
ood-tail	task-label	0.007	0.014	0.002	0.006	0.000	1
ood-tail	uncertainty	-0.028	0.054	0.011	0.027	0.000	0
rare-combo	replay	-0.021	0.041	-0.017	0.014	-0.007	0
rare-combo	cal-risk	-0.021	0.041	0.010	0.131	-0.007	0
rare-combo	fail-active	-0.021	0.041	0.011	0.132	-0.007	0
rare-combo	fs-v4	0.000	0.000	0.003	0.050	0.000	0
rare-combo	hgb-active	-0.021	0.041	-0.018	0.019	-0.007	0
rare-combo	oracle-success	-0.021	0.041	0.020	0.100	-0.007	0
rare-combo	hybrid	-0.021	0.041	-0.001	0.042	-0.007	0
rare-combo	oracle-strata	-0.021	0.041	0.020	0.081	-0.007	0
rare-combo	rf-active	-0.021	0.041	0.014	0.063	-0.007	0
rare-combo	random	-0.021	0.041	-0.008	0.027	-0.007	0

Table 9 continued

Split	Comparison	Succ diff	CI	Macro diff	Rare diff	Safety red.	Better seeds
rare-combo	coreset	0.021	0.096	0.015	0.080	-0.007	1
rare-combo	tail-risk	-0.021	0.041	-0.020	0.065	-0.007	0
rare-combo	task-label	0.000	0.000	0.013	0.082	-0.007	0
rare-combo	uncertainty	-0.021	0.041	-0.001	0.046	-0.007	0
rare-slip	replay	0.000	0.000	0.014	0.045	0.000	0
rare-slip	cal-risk	0.000	0.000	0.061	0.037	0.000	0
rare-slip	fail-active	0.000	0.000	0.061	0.037	0.000	0
rare-slip	fs-v4	0.000	0.000	0.011	0.045	0.000	0
rare-slip	hgb-active	0.000	0.000	0.012	0.056	0.000	0
rare-slip	oracle-success	0.000	0.000	0.020	0.021	0.000	0
rare-slip	hybrid	0.000	0.000	0.009	0.016	0.000	0
rare-slip	oracle-strata	0.000	0.000	-0.011	-0.009	0.000	0
rare-slip	rf-active	0.000	0.000	0.022	0.104	0.000	0
rare-slip	random	0.000	0.000	0.008	0.062	0.000	0
rare-slip	coreset	0.007	0.014	0.004	0.022	0.007	1
rare-slip	tail-risk	0.000	0.000	0.022	0.043	0.000	0
rare-slip	task-label	0.000	0.000	0.002	0.049	0.000	0
rare-slip	uncertainty	0.000	0.000	-0.010	0.026	0.000	0

C AGGREGATE PAIRWISE COMPARISONS

Table 10: Aggregate paired comparisons. These rows drive the hard-regime and combined/extreme lower-bound gates.

Split	Comparison	Succ diff	CI	Macro diff	Rare diff	Safety red.	Better seeds
comb/extreme	replay	-0.012	0.023	-0.008	-0.000	-0.002	0
comb/extreme	cal-risk	-0.012	0.023	0.003	0.071	-0.002	0
comb/extreme	fail-active	-0.012	0.023	0.004	0.068	-0.002	0
comb/extreme	fs-v4	0.000	0.000	0.003	0.008	0.000	0
comb/extreme	hgb-active	-0.012	0.023	-0.008	-0.006	-0.002	0
comb/extreme	oracle-success	-0.012	0.023	0.011	0.013	-0.002	0
comb/extreme	hybrid	-0.012	0.023	-0.002	0.013	-0.002	0
comb/extreme	oracle-strata	-0.012	0.023	0.013	0.018	-0.002	0
comb/extreme	rf-active	-0.012	0.023	0.002	0.005	-0.002	0
comb/extreme	random	-0.012	0.023	-0.003	0.009	-0.002	0
comb/extreme	coreset	0.007	0.045	0.008	0.015	-0.007	1
comb/extreme	tail-risk	-0.012	0.023	-0.006	0.041	-0.002	0
comb/extreme	task-label	0.005	0.009	0.003	0.021	-0.007	1
comb/extreme	uncertainty	-0.012	0.023	-0.001	0.020	-0.002	0
hard	replay	-0.008	0.016	-0.006	-0.008	-0.006	0
hard	cal-risk	-0.008	0.016	0.003	0.032	-0.006	0
hard	fail-active	-0.008	0.016	0.003	0.031	-0.006	0
hard	fs-v4	0.000	0.000	-0.000	-0.018	0.000	0
hard	hgb-active	-0.008	0.016	-0.007	-0.020	-0.006	0
hard	oracle-success	-0.008	0.016	-0.001	-0.048	-0.006	0
hard	hybrid	-0.008	0.016	-0.005	-0.005	-0.006	0
hard	oracle-strata	-0.008	0.016	0.002	-0.039	-0.006	0
hard	rf-active	-0.008	0.016	0.003	-0.003	-0.006	0
hard	random	-0.008	0.016	-0.005	-0.012	-0.006	0
hard	coreset	0.004	0.031	0.002	-0.029	-0.010	1
hard	tail-risk	-0.008	0.016	-0.009	0.009	-0.006	0
hard	task-label	0.001	0.003	-0.001	-0.005	-0.019	1
hard	uncertainty	-0.008	0.016	-0.001	-0.002	-0.006	0

D FIXED-RISK APPENDIX

Table 11: All fixed-risk summaries over all splits, aggregate splits, and risk budgets.

Method	Split	Budget	Success	Coverage	Safety	Tail
replay	actuator	0.05	0.542 $\pm$ 0.133	0.542	0.000	0.000
replay	actuator	0.10	0.903 $\pm$ 0.067	0.903	0.000	0.000
replay	actuator	0.15	0.993 $\pm$ 0.014	0.993	0.000	0.000
replay	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
replay	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
replay	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	comb/extreme	0.15	0.005 $\pm$ 0.009	0.005	0.000	0.000
replay	comb/extreme	0.20	0.012 $\pm$ 0.012	0.012	0.000	0.000
replay	comb/extreme	0.30	0.049 $\pm$ 0.027	0.058	0.002	0.009
replay	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	combined	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
replay	combined	0.20	0.021 $\pm$ 0.029	0.021	0.000	0.000
replay	combined	0.30	0.111 $\pm$ 0.071	0.125	0.007	0.014
replay	sensor-actuator	0.05	0.007 $\pm$ 0.014	0.021	0.007	0.014
replay	sensor-actuator	0.10	0.111 $\pm$ 0.087	0.208	0.062	0.097
replay	sensor-actuator	0.15	0.375 $\pm$ 0.151	0.528	0.097	0.153
replay	sensor-actuator	0.20	0.507 $\pm$ 0.114	0.722	0.153	0.215
replay	sensor-actuator	0.30	0.667 $\pm$ 0.082	0.910	0.167	0.243
replay	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	hard	0.05	0.001 $\pm$ 0.003	0.004	0.001	0.003
replay	hard	0.10	0.022 $\pm$ 0.017	0.042	0.013	0.019
replay	hard	0.15	0.078 $\pm$ 0.032	0.108	0.019	0.031
replay	hard	0.20	0.108 $\pm$ 0.027	0.151	0.031	0.043
replay	hard	0.30	0.163 $\pm$ 0.028	0.217	0.035	0.054
replay	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	nominal	0.05	0.993 $\pm$ 0.014	0.993	0.000	0.000
replay	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
replay	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
replay	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
replay	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
replay	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
replay	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	ood-tail	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
replay	rare-combo	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
replay	rare-combo	0.20	0.014 $\pm$ 0.018	0.014	0.000	0.000
replay	rare-combo	0.30	0.035 $\pm$ 0.029	0.049	0.000	0.014
replay	rare-slip	0.05	0.333 $\pm$ 0.184	0.340	0.000	0.007
replay	rare-slip	0.10	0.715 $\pm$ 0.146	0.729	0.007	0.014
replay	rare-slip	0.15	0.875 $\pm$ 0.082	0.896	0.007	0.021
replay	rare-slip	0.20	0.910 $\pm$ 0.065	0.938	0.007	0.028
replay	rare-slip	0.30	0.951 $\pm$ 0.032	0.986	0.007	0.035
cal-risk	actuator	0.05	0.604 $\pm$ 0.104	0.604	0.000	0.000
cal-risk	actuator	0.10	0.917 $\pm$ 0.046	0.917	0.000	0.000
cal-risk	actuator	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	comb/extreme	0.15	0.012 $\pm$ 0.012	0.012	0.000	0.000
cal-risk	comb/extreme	0.20	0.021 $\pm$ 0.013	0.023	0.000	0.002
cal-risk	comb/extreme	0.30	0.090 $\pm$ 0.042	0.116	0.002	0.025
cal-risk	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	combined	0.15	0.028 $\pm$ 0.029	0.028	0.000	0.000
cal-risk	combined	0.20	0.049 $\pm$ 0.038	0.056	0.000	0.007
cal-risk	combined	0.30	0.174 $\pm$ 0.112	0.188	0.007	0.014
cal-risk	sensor-actuator	0.05	0.014 $\pm$ 0.018	0.028	0.007	0.014
cal-risk	sensor-actuator	0.10	0.139 $\pm$ 0.085	0.236	0.056	0.097
cal-risk	sensor-actuator	0.15	0.354 $\pm$ 0.138	0.521	0.111	0.167
cal-risk	sensor-actuator	0.20	0.521 $\pm$ 0.114	0.729	0.146	0.208
cal-risk	sensor-actuator	0.30	0.667 $\pm$ 0.074	0.910	0.167	0.243
cal-risk	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	hard	0.05	0.003 $\pm$ 0.004	0.006	0.001	0.003
cal-risk	hard	0.10	0.028 $\pm$ 0.017	0.047	0.011	0.019
cal-risk	hard	0.15	0.078 $\pm$ 0.030	0.111	0.022	0.033
cal-risk	hard	0.20	0.117 $\pm$ 0.029	0.160	0.029	0.043

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
cal-risk	hard	0.30	0.188 $\pm$ 0.033	0.251	0.035	0.064
cal-risk	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	nominal	0.05	0.993 $\pm$ 0.014	0.993	0.000	0.000
cal-risk	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
cal-risk	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	ood-tail	0.30	0.007 $\pm$ 0.014	0.028	0.000	0.021
cal-risk	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
cal-risk	rare-combo	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
cal-risk	rare-combo	0.20	0.014 $\pm$ 0.018	0.014	0.000	0.000
cal-risk	rare-combo	0.30	0.090 $\pm$ 0.077	0.132	0.000	0.042
cal-risk	rare-slip	0.05	0.347 $\pm$ 0.178	0.354	0.000	0.007
cal-risk	rare-slip	0.10	0.729 $\pm$ 0.126	0.743	0.007	0.014
cal-risk	rare-slip	0.15	0.861 $\pm$ 0.077	0.889	0.007	0.028
cal-risk	rare-slip	0.20	0.896 $\pm$ 0.063	0.924	0.007	0.028
cal-risk	rare-slip	0.30	0.944 $\pm$ 0.036	0.979	0.007	0.035
fail-active	actuator	0.05	0.604 $\pm$ 0.102	0.604	0.000	0.000
fail-active	actuator	0.10	0.910 $\pm$ 0.050	0.910	0.000	0.000
fail-active	actuator	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
fail-active	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
fail-active	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
fail-active	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fail-active	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fail-active	comb/extreme	0.15	0.012 $\pm$ 0.012	0.012	0.000	0.000
fail-active	comb/extreme	0.20	0.021 $\pm$ 0.013	0.023	0.000	0.002
fail-active	comb/extreme	0.30	0.097 $\pm$ 0.042	0.123	0.002	0.025
fail-active	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fail-active	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fail-active	combined	0.15	0.028 $\pm$ 0.029	0.028	0.000	0.000
fail-active	combined	0.20	0.049 $\pm$ 0.038	0.056	0.000	0.007
fail-active	combined	0.30	0.174 $\pm$ 0.112	0.188	0.007	0.014
fail-active	sensor-actuator	0.05	0.014 $\pm$ 0.018	0.028	0.007	0.014

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
fail-active	sensor-actuator	0.10	0.139 ± 0.085	0.243	0.062	0.104
fail-active	sensor-actuator	0.15	0.340 ± 0.148	0.514	0.118	0.174
fail-active	sensor-actuator	0.20	0.507 ± 0.126	0.715	0.146	0.208
fail-active	sensor-actuator	0.30	0.667 ± 0.074	0.910	0.167	0.243
fail-active	fixture-shift	0.05	0.000 ± 0.000	0.000	0.000	0.000
fail-active	fixture-shift	0.10	0.000 ± 0.000	0.000	0.000	0.000
fail-active	fixture-shift	0.15	0.000 ± 0.000	0.000	0.000	0.000
fail-active	fixture-shift	0.20	0.000 ± 0.000	0.000	0.000	0.000
fail-active	fixture-shift	0.30	0.000 ± 0.000	0.000	0.000	0.000
fail-active	hard	0.05	0.003 ± 0.004	0.006	0.001	0.003
fail-active	hard	0.10	0.028 ± 0.017	0.049	0.013	0.021
fail-active	hard	0.15	0.075 ± 0.032	0.110	0.024	0.035
fail-active	hard	0.20	0.114 ± 0.032	0.157	0.029	0.043
fail-active	hard	0.30	0.192 ± 0.034	0.256	0.035	0.064
fail-active	jammed	0.05	0.000 ± 0.000	0.000	0.000	0.000
fail-active	jammed	0.10	0.000 ± 0.000	0.000	0.000	0.000
fail-active	jammed	0.15	0.000 ± 0.000	0.000	0.000	0.000
fail-active	jammed	0.20	0.000 ± 0.000	0.000	0.000	0.000
fail-active	jammed	0.30	0.000 ± 0.000	0.000	0.000	0.000
fail-active	nominal	0.05	0.993 ± 0.014	0.993	0.000	0.000
fail-active	nominal	0.10	1.000 ± 0.000	1.000	0.000	0.000
fail-active	nominal	0.15	1.000 ± 0.000	1.000	0.000	0.000
fail-active	nominal	0.20	1.000 ± 0.000	1.000	0.000	0.000
fail-active	nominal	0.30	1.000 ± 0.000	1.000	0.000	0.000
fail-active	ood-tail	0.05	0.000 ± 0.000	0.000	0.000	0.000
fail-active	ood-tail	0.10	0.000 ± 0.000	0.000	0.000	0.000
fail-active	ood-tail	0.15	0.000 ± 0.000	0.000	0.000	0.000
fail-active	ood-tail	0.20	0.000 ± 0.000	0.000	0.000	0.000
fail-active	ood-tail	0.30	0.007 ± 0.014	0.028	0.000	0.021
fail-active	rare-combo	0.05	0.000 ± 0.000	0.000	0.000	0.000
fail-active	rare-combo	0.10	0.000 ± 0.000	0.000	0.000	0.000
fail-active	rare-combo	0.15	0.007 ± 0.014	0.007	0.000	0.000
fail-active	rare-combo	0.20	0.014 ± 0.018	0.014	0.000	0.000
fail-active	rare-combo	0.30	0.111 ± 0.068	0.153	0.000	0.042
fail-active	rare-slip	0.05	0.347 ± 0.174	0.354	0.000	0.007
fail-active	rare-slip	0.10	0.722 ± 0.123	0.736	0.007	0.014
fail-active	rare-slip	0.15	0.854 ± 0.080	0.882	0.007	0.028
fail-active	rare-slip	0.20	0.896 ± 0.063	0.924	0.007	0.028
fail-active	rare-slip	0.30	0.938 ± 0.038	0.972	0.007	0.035
fs-v4	actuator	0.05	0.507 ± 0.124	0.514	0.000	0.007
fs-v4	actuator	0.10	0.875 ± 0.049	0.882	0.000	0.007
fs-v4	actuator	0.15	0.993 ± 0.014	1.000	0.000	0.007

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
fs-v4	actuator	0.20	0.993 ± 0.014	1.000	0.000	0.007
fs-v4	actuator	0.30	0.993 ± 0.014	1.000	0.000	0.007
fs-v4	comb/extreme	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	comb/extreme	0.10	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	comb/extreme	0.15	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	comb/extreme	0.20	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	comb/extreme	0.30	0.012 ± 0.010	0.012	0.000	0.000
fs-v4	combined	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	combined	0.10	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	combined	0.15	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	combined	0.20	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	combined	0.30	0.014 ± 0.018	0.014	0.000	0.000
fs-v4	sensor-actuator	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	sensor-actuator	0.10	0.062 ± 0.063	0.125	0.035	0.062
fs-v4	sensor-actuator	0.15	0.139 ± 0.111	0.243	0.069	0.104
fs-v4	sensor-actuator	0.20	0.257 ± 0.196	0.368	0.069	0.111
fs-v4	sensor-actuator	0.30	0.382 ± 0.226	0.528	0.104	0.146
fs-v4	fixture-shift	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	fixture-shift	0.10	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	fixture-shift	0.15	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	fixture-shift	0.20	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	fixture-shift	0.30	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	hard	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	hard	0.10	0.013 ± 0.013	0.025	0.007	0.013
fs-v4	hard	0.15	0.028 ± 0.022	0.049	0.014	0.021
fs-v4	hard	0.20	0.051 ± 0.039	0.074	0.014	0.022
fs-v4	hard	0.30	0.083 ± 0.048	0.113	0.021	0.029
fs-v4	jammed	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	jammed	0.10	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	jammed	0.15	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	jammed	0.20	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	jammed	0.30	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	nominal	0.05	0.993 ± 0.014	0.993	0.000	0.000
fs-v4	nominal	0.10	1.000 ± 0.000	1.000	0.000	0.000
fs-v4	nominal	0.15	1.000 ± 0.000	1.000	0.000	0.000
fs-v4	nominal	0.20	1.000 ± 0.000	1.000	0.000	0.000
fs-v4	nominal	0.30	1.000 ± 0.000	1.000	0.000	0.000
fs-v4	ood-tail	0.05	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	ood-tail	0.10	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	ood-tail	0.15	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	ood-tail	0.20	0.000 ± 0.000	0.000	0.000	0.000
fs-v4	ood-tail	0.30	0.000 ± 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
fs-v4	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v4	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v4	rare-combo	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v4	rare-combo	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v4	rare-combo	0.30	0.021 $\pm$ 0.020	0.021	0.000	0.000
fs-v4	rare-slip	0.05	0.215 $\pm$ 0.102	0.215	0.000	0.000
fs-v4	rare-slip	0.10	0.681 $\pm$ 0.139	0.694	0.007	0.014
fs-v4	rare-slip	0.15	0.854 $\pm$ 0.062	0.875	0.007	0.021
fs-v4	rare-slip	0.20	0.896 $\pm$ 0.060	0.917	0.007	0.021
fs-v4	rare-slip	0.30	0.944 $\pm$ 0.036	0.979	0.007	0.035
fs-v5	actuator	0.05	0.535 $\pm$ 0.122	0.542	0.000	0.007
fs-v5	actuator	0.10	0.889 $\pm$ 0.074	0.896	0.000	0.007
fs-v5	actuator	0.15	0.972 $\pm$ 0.029	0.979	0.000	0.007
fs-v5	actuator	0.20	0.993 $\pm$ 0.014	1.000	0.000	0.007
fs-v5	actuator	0.30	0.993 $\pm$ 0.014	1.000	0.000	0.007
fs-v5	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	comb/extreme	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	comb/extreme	0.20	0.007 $\pm$ 0.010	0.007	0.000	0.000
fs-v5	comb/extreme	0.30	0.039 $\pm$ 0.017	0.049	0.002	0.009
fs-v5	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	combined	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	combined	0.20	0.014 $\pm$ 0.018	0.014	0.000	0.000
fs-v5	combined	0.30	0.083 $\pm$ 0.054	0.097	0.007	0.014
fs-v5	sensor-actuator	0.05	0.000 $\pm$ 0.000	0.014	0.007	0.014
fs-v5	sensor-actuator	0.10	0.104 $\pm$ 0.081	0.188	0.062	0.083
fs-v5	sensor-actuator	0.15	0.340 $\pm$ 0.131	0.493	0.104	0.153
fs-v5	sensor-actuator	0.20	0.507 $\pm$ 0.110	0.715	0.160	0.208
fs-v5	sensor-actuator	0.30	0.674 $\pm$ 0.078	0.924	0.181	0.250
fs-v5	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	hard	0.05	0.000 $\pm$ 0.000	0.003	0.001	0.003
fs-v5	hard	0.10	0.021 $\pm$ 0.016	0.037	0.013	0.017
fs-v5	hard	0.15	0.068 $\pm$ 0.026	0.099	0.021	0.031
fs-v5	hard	0.20	0.106 $\pm$ 0.026	0.147	0.032	0.042
fs-v5	hard	0.30	0.158 $\pm$ 0.024	0.214	0.037	0.056
fs-v5	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
fs-v5	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	nominal	0.05	0.986 $\pm$ 0.018	0.986	0.000	0.000
fs-v5	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
fs-v5	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
fs-v5	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
fs-v5	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
fs-v5	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	ood-tail	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	rare-combo	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
fs-v5	rare-combo	0.20	0.007 $\pm$ 0.014	0.007	0.000	0.000
fs-v5	rare-combo	0.30	0.035 $\pm$ 0.020	0.049	0.000	0.014
fs-v5	rare-slip	0.05	0.306 $\pm$ 0.175	0.306	0.000	0.000
fs-v5	rare-slip	0.10	0.708 $\pm$ 0.145	0.722	0.007	0.014
fs-v5	rare-slip	0.15	0.861 $\pm$ 0.080	0.882	0.007	0.021
fs-v5	rare-slip	0.20	0.917 $\pm$ 0.065	0.938	0.007	0.021
fs-v5	rare-slip	0.30	0.951 $\pm$ 0.032	0.986	0.007	0.035
hgb-active	actuator	0.05	0.576 $\pm$ 0.144	0.576	0.000	0.000
hgb-active	actuator	0.10	0.875 $\pm$ 0.091	0.875	0.000	0.000
hgb-active	actuator	0.15	0.993 $\pm$ 0.014	0.993	0.000	0.000
hgb-active	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
hgb-active	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
hgb-active	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	comb/extreme	0.15	0.005 $\pm$ 0.009	0.005	0.000	0.000
hgb-active	comb/extreme	0.20	0.012 $\pm$ 0.012	0.012	0.000	0.000
hgb-active	comb/extreme	0.30	0.042 $\pm$ 0.029	0.053	0.002	0.012
hgb-active	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	combined	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
hgb-active	combined	0.20	0.028 $\pm$ 0.029	0.028	0.000	0.000
hgb-active	combined	0.30	0.104 $\pm$ 0.078	0.125	0.007	0.021
hgb-active	sensor-actuator	0.05	0.007 $\pm$ 0.014	0.021	0.007	0.014
hgb-active	sensor-actuator	0.10	0.132 $\pm$ 0.087	0.215	0.049	0.083
hgb-active	sensor-actuator	0.15	0.299 $\pm$ 0.148	0.438	0.097	0.139
hgb-active	sensor-actuator	0.20	0.451 $\pm$ 0.131	0.639	0.132	0.188

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
hgb-active	sensor-actuator	0.30	0.625 $\pm$ 0.096	0.861	0.167	0.236
hgb-active	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	hard	0.05	0.001 $\pm$ 0.003	0.004	0.001	0.003
hgb-active	hard	0.10	0.026 $\pm$ 0.017	0.043	0.010	0.017
hgb-active	hard	0.15	0.062 $\pm$ 0.032	0.090	0.019	0.028
hgb-active	hard	0.20	0.097 $\pm$ 0.030	0.135	0.026	0.037
hgb-active	hard	0.30	0.150 $\pm$ 0.034	0.204	0.035	0.054
hgb-active	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	nominal	0.05	0.986 $\pm$ 0.018	0.986	0.000	0.000
hgb-active	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
hgb-active	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
hgb-active	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
hgb-active	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
hgb-active	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	ood-tail	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hgb-active	rare-combo	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
hgb-active	rare-combo	0.20	0.007 $\pm$ 0.014	0.007	0.000	0.000
hgb-active	rare-combo	0.30	0.021 $\pm$ 0.020	0.035	0.000	0.014
hgb-active	rare-slip	0.05	0.333 $\pm$ 0.166	0.340	0.000	0.007
hgb-active	rare-slip	0.10	0.722 $\pm$ 0.138	0.736	0.007	0.014
hgb-active	rare-slip	0.15	0.854 $\pm$ 0.080	0.882	0.007	0.028
hgb-active	rare-slip	0.20	0.896 $\pm$ 0.063	0.924	0.007	0.028
hgb-active	rare-slip	0.30	0.951 $\pm$ 0.032	0.986	0.007	0.035
oracle-success	actuator	0.05	0.500 $\pm$ 0.130	0.500	0.000	0.000
oracle-success	actuator	0.10	0.854 $\pm$ 0.085	0.854	0.000	0.000
oracle-success	actuator	0.15	0.972 $\pm$ 0.041	0.972	0.000	0.000
oracle-success	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-success	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-success	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
oracle-success	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	comb/extreme	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	comb/extreme	0.20	0.005 $\pm$ 0.009	0.005	0.000	0.000
oracle-success	comb/extreme	0.30	0.037 $\pm$ 0.017	0.053	0.000	0.016
oracle-success	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	combined	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	combined	0.20	0.007 $\pm$ 0.014	0.007	0.000	0.000
oracle-success	combined	0.30	0.069 $\pm$ 0.049	0.076	0.000	0.007
oracle-success	sensor-actuator	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	sensor-actuator	0.10	0.007 $\pm$ 0.014	0.014	0.007	0.007
oracle-success	sensor-actuator	0.15	0.035 $\pm$ 0.054	0.056	0.021	0.021
oracle-success	sensor-actuator	0.20	0.083 $\pm$ 0.090	0.125	0.028	0.042
oracle-success	sensor-actuator	0.30	0.243 $\pm$ 0.116	0.375	0.097	0.132
oracle-success	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	hard	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	hard	0.10	0.001 $\pm$ 0.003	0.003	0.001	0.001
oracle-success	hard	0.15	0.007 $\pm$ 0.011	0.011	0.004	0.004
oracle-success	hard	0.20	0.019 $\pm$ 0.023	0.028	0.006	0.008
oracle-success	hard	0.30	0.071 $\pm$ 0.025	0.107	0.019	0.036
oracle-success	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	nominal	0.05	0.972 $\pm$ 0.041	0.972	0.000	0.000
oracle-success	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-success	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-success	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-success	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-success	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	ood-tail	0.30	0.007 $\pm$ 0.014	0.035	0.000	0.028
oracle-success	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-success	rare-combo	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
oracle-success	rare-combo	0.20	0.007 $\pm$ 0.014	0.007	0.000	0.000
oracle-success	rare-combo	0.30	0.035 $\pm$ 0.020	0.049	0.000	0.014
oracle-success	rare-slip	0.05	0.250 $\pm$ 0.109	0.257	0.000	0.007
oracle-success	rare-slip	0.10	0.681 $\pm$ 0.125	0.701	0.007	0.021
oracle-success	rare-slip	0.15	0.833 $\pm$ 0.058	0.854	0.007	0.021
oracle-success	rare-slip	0.20	0.889 $\pm$ 0.065	0.917	0.007	0.028
oracle-success	rare-slip	0.30	0.944 $\pm$ 0.036	0.979	0.007	0.035
hybrid	actuator	0.05	0.583 $\pm$ 0.109	0.583	0.000	0.000
hybrid	actuator	0.10	0.903 $\pm$ 0.057	0.903	0.000	0.000
hybrid	actuator	0.15	0.993 $\pm$ 0.014	0.993	0.000	0.000
hybrid	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
hybrid	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
hybrid	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	comb/extreme	0.15	0.002 $\pm$ 0.005	0.002	0.000	0.000
hybrid	comb/extreme	0.20	0.016 $\pm$ 0.014	0.019	0.000	0.002
hybrid	comb/extreme	0.30	0.053 $\pm$ 0.027	0.086	0.002	0.032
hybrid	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	combined	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
hybrid	combined	0.20	0.028 $\pm$ 0.041	0.035	0.000	0.007
hybrid	combined	0.30	0.118 $\pm$ 0.083	0.139	0.007	0.021
hybrid	sensor-actuator	0.05	0.000 $\pm$ 0.000	0.007	0.000	0.007
hybrid	sensor-actuator	0.10	0.160 $\pm$ 0.081	0.236	0.049	0.076
hybrid	sensor-actuator	0.15	0.375 $\pm$ 0.164	0.521	0.097	0.146
hybrid	sensor-actuator	0.20	0.549 $\pm$ 0.122	0.757	0.139	0.208
hybrid	sensor-actuator	0.30	0.674 $\pm$ 0.070	0.938	0.167	0.264
hybrid	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	hard	0.05	0.000 $\pm$ 0.000	0.001	0.000	0.001
hybrid	hard	0.10	0.032 $\pm$ 0.016	0.047	0.010	0.015
hybrid	hard	0.15	0.076 $\pm$ 0.034	0.106	0.019	0.029
hybrid	hard	0.20	0.119 $\pm$ 0.033	0.163	0.028	0.043
hybrid	hard	0.30	0.167 $\pm$ 0.025	0.239	0.035	0.072
hybrid	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
hybrid	nominal	0.05	0.993 $\pm$ 0.014	0.993	0.000	0.000
hybrid	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
hybrid	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
hybrid	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
hybrid	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
hybrid	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	ood-tail	0.30	0.000 $\pm$ 0.000	0.035	0.000	0.035
hybrid	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	rare-combo	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
hybrid	rare-combo	0.20	0.021 $\pm$ 0.020	0.021	0.000	0.000
hybrid	rare-combo	0.30	0.042 $\pm$ 0.018	0.083	0.000	0.042
hybrid	rare-slip	0.05	0.389 $\pm$ 0.192	0.396	0.000	0.007
hybrid	rare-slip	0.10	0.736 $\pm$ 0.147	0.750	0.007	0.014
hybrid	rare-slip	0.15	0.882 $\pm$ 0.073	0.910	0.007	0.028
hybrid	rare-slip	0.20	0.903 $\pm$ 0.064	0.938	0.007	0.035
hybrid	rare-slip	0.30	0.958 $\pm$ 0.027	0.993	0.007	0.035
oracle-strata	actuator	0.05	0.493 $\pm$ 0.128	0.493	0.000	0.000
oracle-strata	actuator	0.10	0.854 $\pm$ 0.092	0.854	0.000	0.000
oracle-strata	actuator	0.15	0.958 $\pm$ 0.040	0.958	0.000	0.000
oracle-strata	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-strata	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
oracle-strata	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	comb/extreme	0.15	0.002 $\pm$ 0.005	0.002	0.000	0.000
oracle-strata	comb/extreme	0.20	0.012 $\pm$ 0.012	0.012	0.000	0.000
oracle-strata	comb/extreme	0.30	0.044 $\pm$ 0.019	0.065	0.000	0.021
oracle-strata	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	combined	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	combined	0.20	0.021 $\pm$ 0.029	0.021	0.000	0.000
oracle-strata	combined	0.30	0.076 $\pm$ 0.054	0.083	0.000	0.007
oracle-strata	sensor-actuator	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	sensor-actuator	0.10	0.014 $\pm$ 0.027	0.021	0.007	0.007
oracle-strata	sensor-actuator	0.15	0.035 $\pm$ 0.054	0.062	0.021	0.028
oracle-strata	sensor-actuator	0.20	0.090 $\pm$ 0.087	0.139	0.035	0.049
oracle-strata	sensor-actuator	0.30	0.278 $\pm$ 0.111	0.424	0.111	0.146
oracle-strata	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
oracle-strata	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
oracle-strata fixture-shift		0.15	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata fixture-shift		0.20	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata fixture-shift		0.30	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata hard		0.05	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata hard		0.10	0.003 ± 0.005	0.004	0.001	0.001
oracle-strata hard		0.15	0.008 ± 0.013	0.014	0.004	0.006
oracle-strata hard		0.20	0.025 ± 0.022	0.035	0.007	0.010
oracle-strata hard		0.30	0.082 ± 0.026	0.124	0.022	0.042
oracle-strata jammed		0.05	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata jammed		0.10	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata jammed		0.15	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata jammed		0.20	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata jammed		0.30	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata nominal		0.05	0.938 ± 0.086	0.938	0.000	0.000
oracle-strata nominal		0.10	1.000 ± 0.000	1.000	0.000	0.000
oracle-strata nominal		0.15	1.000 ± 0.000	1.000	0.000	0.000
oracle-strata nominal		0.20	1.000 ± 0.000	1.000	0.000	0.000
oracle-strata nominal		0.30	1.000 ± 0.000	1.000	0.000	0.000
oracle-strata ood-tail		0.05	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata ood-tail		0.10	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata ood-tail		0.15	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata ood-tail		0.20	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata ood-tail		0.30	0.007 ± 0.014	0.035	0.000	0.028
oracle-strata rare-combo		0.05	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata rare-combo		0.10	0.000 ± 0.000	0.000	0.000	0.000
oracle-strata rare-combo		0.15	0.007 ± 0.014	0.007	0.000	0.000
oracle-strata rare-combo		0.20	0.014 ± 0.018	0.014	0.000	0.000
oracle-strata rare-combo		0.30	0.049 ± 0.025	0.076	0.000	0.028
oracle-strata rare-slip		0.05	0.160 ± 0.095	0.160	0.000	0.000
oracle-strata rare-slip		0.10	0.521 ± 0.171	0.535	0.007	0.014
oracle-strata rare-slip		0.15	0.750 ± 0.116	0.771	0.007	0.021
oracle-strata rare-slip		0.20	0.840 ± 0.063	0.861	0.007	0.021
oracle-strata rare-slip		0.30	0.882 ± 0.066	0.917	0.007	0.035
rf-active actuator		0.05	0.528 ± 0.122	0.528	0.000	0.000
rf-active actuator		0.10	0.833 ± 0.068	0.833	0.000	0.000
rf-active actuator		0.15	0.958 ± 0.040	0.958	0.000	0.000
rf-active actuator		0.20	1.000 ± 0.000	1.000	0.000	0.000
rf-active actuator		0.30	1.000 ± 0.000	1.000	0.000	0.000
rf-active comb/extreme		0.05	0.000 ± 0.000	0.000	0.000	0.000
rf-active comb/extreme		0.10	0.002 ± 0.005	0.002	0.000	0.000
rf-active comb/extreme		0.15	0.012 ± 0.010	0.012	0.000	0.000
rf-active comb/extreme		0.20	0.023 ± 0.013	0.025	0.000	0.002

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
rf-active comb/extreme		0.30	0.081 ± 0.026	0.111	0.005	0.030
rf-active combined		0.05	0.000 ± 0.000	0.000	0.000	0.000
rf-active combined		0.10	0.000 ± 0.000	0.000	0.000	0.000
rf-active combined		0.15	0.014 ± 0.018	0.014	0.000	0.000
rf-active combined		0.20	0.049 ± 0.032	0.056	0.000	0.007
rf-active combined		0.30	0.118 ± 0.066	0.139	0.007	0.021
rf-active sensor-actuator		0.05	0.049 ± 0.048	0.083	0.021	0.035
rf-active sensor-actuator		0.10	0.201 ± 0.109	0.312	0.069	0.111
rf-active sensor-actuator		0.15	0.396 ± 0.160	0.576	0.132	0.181
rf-active sensor-actuator		0.20	0.542 ± 0.123	0.764	0.160	0.222
rf-active sensor-actuator		0.30	0.667 ± 0.074	0.910	0.167	0.243
rf-active fixture-shift		0.05	0.000 ± 0.000	0.000	0.000	0.000
rf-active fixture-shift		0.10	0.000 ± 0.000	0.000	0.000	0.000
rf-active fixture-shift		0.15	0.000 ± 0.000	0.000	0.000	0.000
rf-active fixture-shift		0.20	0.000 ± 0.000	0.000	0.000	0.000
rf-active fixture-shift		0.30	0.000 ± 0.000	0.000	0.000	0.000
rf-active hard		0.05	0.010 ± 0.010	0.017	0.004	0.007
rf-active hard		0.10	0.042 ± 0.022	0.064	0.014	0.022
rf-active hard		0.15	0.086 ± 0.035	0.122	0.026	0.036
rf-active hard		0.20	0.122 ± 0.031	0.168	0.032	0.046
rf-active hard		0.30	0.182 ± 0.026	0.249	0.036	0.067
rf-active jammed		0.05	0.000 ± 0.000	0.000	0.000	0.000
rf-active jammed		0.10	0.000 ± 0.000	0.000	0.000	0.000
rf-active jammed		0.15	0.000 ± 0.000	0.000	0.000	0.000
rf-active jammed		0.20	0.000 ± 0.000	0.000	0.000	0.000
rf-active jammed		0.30	0.000 ± 0.000	0.000	0.000	0.000
rf-active nominal		0.05	0.972 ± 0.036	0.972	0.000	0.000
rf-active nominal		0.10	1.000 ± 0.000	1.000	0.000	0.000
rf-active nominal		0.15	1.000 ± 0.000	1.000	0.000	0.000
rf-active nominal		0.20	1.000 ± 0.000	1.000	0.000	0.000
rf-active nominal		0.30	1.000 ± 0.000	1.000	0.000	0.000
rf-active ood-tail		0.05	0.000 ± 0.000	0.000	0.000	0.000
rf-active ood-tail		0.10	0.000 ± 0.000	0.000	0.000	0.000
rf-active ood-tail		0.15	0.000 ± 0.000	0.000	0.000	0.000
rf-active ood-tail		0.20	0.000 ± 0.000	0.000	0.000	0.000
rf-active ood-tail		0.30	0.000 ± 0.000	0.007	0.000	0.007
rf-active rare-combo		0.05	0.000 ± 0.000	0.000	0.000	0.000
rf-active rare-combo		0.10	0.007 ± 0.014	0.007	0.000	0.000
rf-active rare-combo		0.15	0.021 ± 0.020	0.021	0.000	0.000
rf-active rare-combo		0.20	0.021 ± 0.020	0.021	0.000	0.000
rf-active rare-combo		0.30	0.125 ± 0.057	0.188	0.007	0.062
rf-active rare-slip		0.05	0.368 ± 0.163	0.375	0.000	0.007

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
rf-active	rare-slip	0.10	0.715 ± 0.136	0.736	0.007	0.021
rf-active	rare-slip	0.15	0.833 ± 0.082	0.854	0.007	0.021
rf-active	rare-slip	0.20	0.889 ± 0.065	0.917	0.007	0.028
rf-active	rare-slip	0.30	0.938 ± 0.043	0.972	0.007	0.035
random	actuator	0.05	0.583 ± 0.111	0.583	0.000	0.000
random	actuator	0.10	0.889 ± 0.068	0.889	0.000	0.000
random	actuator	0.15	0.986 ± 0.018	0.986	0.000	0.000
random	actuator	0.20	1.000 ± 0.000	1.000	0.000	0.000
random	actuator	0.30	1.000 ± 0.000	1.000	0.000	0.000
random	comb/extreme	0.05	0.000 ± 0.000	0.000	0.000	0.000
random	comb/extreme	0.10	0.000 ± 0.000	0.000	0.000	0.000
random	comb/extreme	0.15	0.012 ± 0.012	0.012	0.000	0.000
random	comb/extreme	0.20	0.016 ± 0.011	0.016	0.000	0.000
random	comb/extreme	0.30	0.060 ± 0.024	0.076	0.002	0.016
random	combined	0.05	0.000 ± 0.000	0.000	0.000	0.000
random	combined	0.10	0.000 ± 0.000	0.000	0.000	0.000
random	combined	0.15	0.021 ± 0.029	0.021	0.000	0.000
random	combined	0.20	0.028 ± 0.029	0.028	0.000	0.000
random	combined	0.30	0.118 ± 0.066	0.139	0.007	0.021
random	sensor-actuator	0.05	0.000 ± 0.000	0.021	0.014	0.021
random	sensor-actuator	0.10	0.111 ± 0.077	0.194	0.049	0.083
random	sensor-actuator	0.15	0.264 ± 0.125	0.382	0.076	0.118
random	sensor-actuator	0.20	0.438 ± 0.153	0.632	0.139	0.194
random	sensor-actuator	0.30	0.625 ± 0.084	0.868	0.167	0.243
random	fixture-shift	0.05	0.000 ± 0.000	0.000	0.000	0.000
random	fixture-shift	0.10	0.000 ± 0.000	0.000	0.000	0.000
random	fixture-shift	0.15	0.000 ± 0.000	0.000	0.000	0.000
random	fixture-shift	0.20	0.000 ± 0.000	0.000	0.000	0.000
random	fixture-shift	0.30	0.000 ± 0.000	0.000	0.000	0.000
random	hard	0.05	0.000 ± 0.000	0.004	0.003	0.004
random	hard	0.10	0.022 ± 0.015	0.039	0.010	0.017
random	hard	0.15	0.060 ± 0.030	0.083	0.015	0.024
random	hard	0.20	0.097 ± 0.036	0.136	0.028	0.039
random	hard	0.30	0.161 ± 0.027	0.219	0.035	0.058
random	jammed	0.05	0.000 ± 0.000	0.000	0.000	0.000
random	jammed	0.10	0.000 ± 0.000	0.000	0.000	0.000
random	jammed	0.15	0.000 ± 0.000	0.000	0.000	0.000
random	jammed	0.20	0.000 ± 0.000	0.000	0.000	0.000
random	jammed	0.30	0.000 ± 0.000	0.000	0.000	0.000
random	nominal	0.05	0.993 ± 0.014	0.993	0.000	0.000
random	nominal	0.10	1.000 ± 0.000	1.000	0.000	0.000
random	nominal	0.15	1.000 ± 0.000	1.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
random	nominal	0.20	1.000 ± 0.000	1.000	0.000	0.000
random	nominal	0.30	1.000 ± 0.000	1.000	0.000	0.000
random	ood-tail	0.05	0.000 ± 0.000	0.000	0.000	0.000
random	ood-tail	0.10	0.000 ± 0.000	0.000	0.000	0.000
random	ood-tail	0.15	0.000 ± 0.000	0.000	0.000	0.000
random	ood-tail	0.20	0.000 ± 0.000	0.000	0.000	0.000
random	ood-tail	0.30	0.000 ± 0.000	0.007	0.000	0.007
random	rare-combo	0.05	0.000 ± 0.000	0.000	0.000	0.000
random	rare-combo	0.10	0.000 ± 0.000	0.000	0.000	0.000
random	rare-combo	0.15	0.014 ± 0.018	0.014	0.000	0.000
random	rare-combo	0.20	0.021 ± 0.020	0.021	0.000	0.000
random	rare-combo	0.30	0.062 ± 0.038	0.083	0.000	0.021
random	rare-slip	0.05	0.340 ± 0.173	0.340	0.000	0.000
random	rare-slip	0.10	0.729 ± 0.132	0.743	0.007	0.014
random	rare-slip	0.15	0.868 ± 0.080	0.896	0.007	0.028
random	rare-slip	0.20	0.896 ± 0.063	0.924	0.007	0.028
random	rare-slip	0.30	0.951 ± 0.032	0.986	0.007	0.035
coreset	actuator	0.05	0.340 ± 0.114	0.340	0.000	0.000
coreset	actuator	0.10	0.778 ± 0.077	0.778	0.000	0.000
coreset	actuator	0.15	0.944 ± 0.046	0.944	0.000	0.000
coreset	actuator	0.20	1.000 ± 0.000	1.000	0.000	0.000
coreset	actuator	0.30	1.000 ± 0.000	1.000	0.000	0.000
coreset	comb/extreme	0.05	0.000 ± 0.000	0.000	0.000	0.000
coreset	comb/extreme	0.10	0.000 ± 0.000	0.000	0.000	0.000
coreset	comb/extreme	0.15	0.000 ± 0.000	0.000	0.000	0.000
coreset	comb/extreme	0.20	0.012 ± 0.012	0.012	0.000	0.000
coreset	comb/extreme	0.30	0.046 ± 0.024	0.053	0.000	0.007
coreset	combined	0.05	0.000 ± 0.000	0.000	0.000	0.000
coreset	combined	0.10	0.000 ± 0.000	0.000	0.000	0.000
coreset	combined	0.15	0.000 ± 0.000	0.000	0.000	0.000
coreset	combined	0.20	0.028 ± 0.029	0.028	0.000	0.000
coreset	combined	0.30	0.104 ± 0.073	0.111	0.000	0.007
coreset	sensor-actuator	0.05	0.000 ± 0.000	0.000	0.000	0.000
coreset	sensor-actuator	0.10	0.007 ± 0.014	0.028	0.014	0.021
coreset	sensor-actuator	0.15	0.076 ± 0.065	0.139	0.035	0.062
coreset	sensor-actuator	0.20	0.215 ± 0.081	0.333	0.076	0.118
coreset	sensor-actuator	0.30	0.493 ± 0.070	0.736	0.160	0.243
coreset	fixture-shift	0.05	0.000 ± 0.000	0.000	0.000	0.000
coreset	fixture-shift	0.10	0.000 ± 0.000	0.000	0.000	0.000
coreset	fixture-shift	0.15	0.000 ± 0.000	0.000	0.000	0.000
coreset	fixture-shift	0.20	0.000 ± 0.000	0.000	0.000	0.000
coreset	fixture-shift	0.30	0.000 ± 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
coreset	hard	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	hard	0.10	0.001 $\pm$ 0.003	0.006	0.003	0.004
coreset	hard	0.15	0.015 $\pm$ 0.013	0.028	0.007	0.013
coreset	hard	0.20	0.050 $\pm$ 0.019	0.074	0.015	0.024
coreset	hard	0.30	0.126 $\pm$ 0.022	0.179	0.032	0.053
coreset	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	nominal	0.05	0.944 $\pm$ 0.046	0.944	0.000	0.000
coreset	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
coreset	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
coreset	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
coreset	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
coreset	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	ood-tail	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	ood-tail	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	rare-combo	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
coreset	rare-combo	0.20	0.007 $\pm$ 0.014	0.007	0.000	0.000
coreset	rare-combo	0.30	0.035 $\pm$ 0.020	0.049	0.000	0.014
coreset	rare-slip	0.05	0.174 $\pm$ 0.114	0.174	0.000	0.000
coreset	rare-slip	0.10	0.611 $\pm$ 0.172	0.632	0.007	0.021
coreset	rare-slip	0.15	0.792 $\pm$ 0.082	0.812	0.007	0.021
coreset	rare-slip	0.20	0.875 $\pm$ 0.064	0.903	0.007	0.028
coreset	rare-slip	0.30	0.944 $\pm$ 0.036	0.986	0.014	0.042
tail-risk	actuator	0.05	0.625 $\pm$ 0.121	0.625	0.000	0.000
tail-risk	actuator	0.10	0.938 $\pm$ 0.038	0.938	0.000	0.000
tail-risk	actuator	0.15	0.993 $\pm$ 0.014	0.993	0.000	0.000
tail-risk	actuator	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
tail-risk	actuator	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
tail-risk	comb/extreme	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	comb/extreme	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	comb/extreme	0.15	0.002 $\pm$ 0.005	0.002	0.000	0.000
tail-risk	comb/extreme	0.20	0.016 $\pm$ 0.011	0.019	0.000	0.002
tail-risk	comb/extreme	0.30	0.067 $\pm$ 0.024	0.109	0.002	0.042
tail-risk	combined	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	combined	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
tail-risk	combined	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	combined	0.20	0.028 $\pm$ 0.029	0.035	0.000	0.007
tail-risk	combined	0.30	0.125 $\pm$ 0.082	0.146	0.007	0.021
tail-risk	sensor-actuator	0.05	0.014 $\pm$ 0.027	0.028	0.007	0.014
tail-risk	sensor-actuator	0.10	0.208 $\pm$ 0.094	0.312	0.062	0.104
tail-risk	sensor-actuator	0.15	0.424 $\pm$ 0.161	0.583	0.104	0.160
tail-risk	sensor-actuator	0.20	0.576 $\pm$ 0.107	0.806	0.153	0.229
tail-risk	sensor-actuator	0.30	0.674 $\pm$ 0.070	0.924	0.167	0.250
tail-risk	fixture-shift	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	fixture-shift	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	fixture-shift	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	fixture-shift	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	fixture-shift	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	hard	0.05	0.003 $\pm$ 0.005	0.006	0.001	0.003
tail-risk	hard	0.10	0.042 $\pm$ 0.019	0.062	0.013	0.021
tail-risk	hard	0.15	0.086 $\pm$ 0.033	0.118	0.021	0.032
tail-risk	hard	0.20	0.125 $\pm$ 0.026	0.172	0.031	0.047
tail-risk	hard	0.30	0.175 $\pm$ 0.025	0.250	0.035	0.075
tail-risk	jammed	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	jammed	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	jammed	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	jammed	0.20	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	jammed	0.30	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	nominal	0.05	0.993 $\pm$ 0.014	0.993	0.000	0.000
tail-risk	nominal	0.10	1.000 $\pm$ 0.000	1.000	0.000	0.000
tail-risk	nominal	0.15	1.000 $\pm$ 0.000	1.000	0.000	0.000
tail-risk	nominal	0.20	1.000 $\pm$ 0.000	1.000	0.000	0.000
tail-risk	nominal	0.30	1.000 $\pm$ 0.000	1.000	0.000	0.000
tail-risk	ood-tail	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	ood-tail	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	ood-tail	0.15	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	ood-tail	0.20	0.007 $\pm$ 0.014	0.007	0.000	0.000
tail-risk	ood-tail	0.30	0.007 $\pm$ 0.014	0.076	0.000	0.069
tail-risk	rare-combo	0.05	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	rare-combo	0.10	0.000 $\pm$ 0.000	0.000	0.000	0.000
tail-risk	rare-combo	0.15	0.007 $\pm$ 0.014	0.007	0.000	0.000
tail-risk	rare-combo	0.20	0.014 $\pm$ 0.018	0.014	0.000	0.000
tail-risk	rare-combo	0.30	0.069 $\pm$ 0.049	0.104	0.000	0.035
tail-risk	rare-slip	0.05	0.417 $\pm$ 0.198	0.431	0.007	0.014
tail-risk	rare-slip	0.10	0.771 $\pm$ 0.104	0.785	0.007	0.014
tail-risk	rare-slip	0.15	0.882 $\pm$ 0.075	0.910	0.007	0.028
tail-risk	rare-slip	0.20	0.917 $\pm$ 0.065	0.944	0.007	0.028

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
tail-risk	rare-slip	0.30	0.958 ± 0.027	0.993	0.007	0.035
task-label	actuator	0.05	0.542 ± 0.106	0.556	0.000	0.014
task-label	actuator	0.10	0.868 ± 0.077	0.882	0.000	0.014
task-label	actuator	0.15	0.986 ± 0.027	1.000	0.000	0.014
task-label	actuator	0.20	0.986 ± 0.027	1.000	0.000	0.014
task-label	actuator	0.30	0.986 ± 0.027	1.000	0.000	0.014
task-label	comb/extreme	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	comb/extreme	0.10	0.000 ± 0.000	0.000	0.000	0.000
task-label	comb/extreme	0.15	0.005 ± 0.009	0.005	0.000	0.000
task-label	comb/extreme	0.20	0.016 ± 0.011	0.019	0.000	0.002
task-label	comb/extreme	0.30	0.056 ± 0.032	0.081	0.002	0.025
task-label	combined	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	combined	0.10	0.000 ± 0.000	0.000	0.000	0.000
task-label	combined	0.15	0.007 ± 0.014	0.007	0.000	0.000
task-label	combined	0.20	0.035 ± 0.029	0.035	0.000	0.000
task-label	combined	0.30	0.118 ± 0.073	0.132	0.007	0.014
task-label	sensor-actuator	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	sensor-actuator	0.10	0.090 ± 0.062	0.167	0.035	0.076
task-label	sensor-actuator	0.15	0.222 ± 0.109	0.361	0.076	0.139
task-label	sensor-actuator	0.20	0.472 ± 0.105	0.674	0.118	0.201
task-label	sensor-actuator	0.30	0.653 ± 0.071	0.889	0.139	0.236
task-label	fixture-shift	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	fixture-shift	0.10	0.000 ± 0.000	0.000	0.000	0.000
task-label	fixture-shift	0.15	0.000 ± 0.000	0.000	0.000	0.000
task-label	fixture-shift	0.20	0.000 ± 0.000	0.000	0.000	0.000
task-label	fixture-shift	0.30	0.000 ± 0.000	0.000	0.000	0.000
task-label	hard	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	hard	0.10	0.018 ± 0.012	0.033	0.007	0.015
task-label	hard	0.15	0.047 ± 0.025	0.075	0.015	0.028
task-label	hard	0.20	0.104 ± 0.025	0.146	0.024	0.042
task-label	hard	0.30	0.164 ± 0.024	0.226	0.029	0.062
task-label	jammed	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	jammed	0.10	0.000 ± 0.000	0.000	0.000	0.000
task-label	jammed	0.15	0.000 ± 0.000	0.000	0.000	0.000
task-label	jammed	0.20	0.000 ± 0.000	0.000	0.000	0.000
task-label	jammed	0.30	0.000 ± 0.000	0.000	0.000	0.000
task-label	nominal	0.05	0.993 ± 0.014	0.993	0.000	0.000
task-label	nominal	0.10	1.000 ± 0.000	1.000	0.000	0.000
task-label	nominal	0.15	1.000 ± 0.000	1.000	0.000	0.000
task-label	nominal	0.20	1.000 ± 0.000	1.000	0.000	0.000
task-label	nominal	0.30	1.000 ± 0.000	1.000	0.000	0.000
task-label	ood-tail	0.05	0.000 ± 0.000	0.000	0.000	0.000

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
task-label	ood-tail	0.10	0.000 ± 0.000	0.000	0.000	0.000
task-label	ood-tail	0.15	0.000 ± 0.000	0.000	0.000	0.000
task-label	ood-tail	0.20	0.000 ± 0.000	0.000	0.000	0.000
task-label	ood-tail	0.30	0.000 ± 0.000	0.021	0.000	0.021
task-label	rare-combo	0.05	0.000 ± 0.000	0.000	0.000	0.000
task-label	rare-combo	0.10	0.000 ± 0.000	0.000	0.000	0.000
task-label	rare-combo	0.15	0.007 ± 0.014	0.007	0.000	0.000
task-label	rare-combo	0.20	0.014 ± 0.018	0.021	0.000	0.007
task-label	rare-combo	0.30	0.049 ± 0.052	0.090	0.000	0.042
task-label	rare-slip	0.05	0.319 ± 0.136	0.319	0.000	0.000
task-label	rare-slip	0.10	0.729 ± 0.137	0.736	0.007	0.007
task-label	rare-slip	0.15	0.854 ± 0.077	0.875	0.007	0.021
task-label	rare-slip	0.20	0.896 ± 0.063	0.917	0.007	0.021
task-label	rare-slip	0.30	0.951 ± 0.032	0.986	0.007	0.035
uncertainty	actuator	0.05	0.597 ± 0.162	0.597	0.000	0.000
uncertainty	actuator	0.10	0.889 ± 0.101	0.889	0.000	0.000
uncertainty	actuator	0.15	0.986 ± 0.018	0.986	0.000	0.000
uncertainty	actuator	0.20	1.000 ± 0.000	1.000	0.000	0.000
uncertainty	actuator	0.30	1.000 ± 0.000	1.000	0.000	0.000
uncertainty	comb/extreme	0.05	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	comb/extreme	0.10	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	comb/extreme	0.15	0.007 ± 0.007	0.007	0.000	0.000
uncertainty	comb/extreme	0.20	0.019 ± 0.012	0.019	0.000	0.000
uncertainty	comb/extreme	0.30	0.065 ± 0.025	0.074	0.002	0.009
uncertainty	combined	0.05	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	combined	0.10	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	combined	0.15	0.014 ± 0.018	0.014	0.000	0.000
uncertainty	combined	0.20	0.035 ± 0.029	0.035	0.000	0.000
uncertainty	combined	0.30	0.146 ± 0.071	0.167	0.007	0.021
uncertainty	sensor-actuator	0.05	0.007 ± 0.014	0.028	0.014	0.021
uncertainty	sensor-actuator	0.10	0.139 ± 0.074	0.250	0.069	0.111
uncertainty	sensor-actuator	0.15	0.375 ± 0.071	0.556	0.125	0.181
uncertainty	sensor-actuator	0.20	0.535 ± 0.090	0.764	0.167	0.229
uncertainty	sensor-actuator	0.30	0.688 ± 0.068	0.938	0.167	0.250
uncertainty	fixture-shift	0.05	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	fixture-shift	0.10	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	fixture-shift	0.15	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	fixture-shift	0.20	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	fixture-shift	0.30	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	hard	0.05	0.001 ± 0.003	0.006	0.003	0.004
uncertainty	hard	0.10	0.028 ± 0.015	0.050	0.014	0.022
uncertainty	hard	0.15	0.079 ± 0.017	0.115	0.025	0.036

Table 11 continued

Method	Split	Budget	Success	Coverage	Safety	Tail
uncertainty	hard	0.20	0.118 ± 0.024	0.164	0.033	0.046
uncertainty	hard	0.30	0.176 ± 0.024	0.232	0.035	0.056
uncertainty	jammed	0.05	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	jammed	0.10	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	jammed	0.15	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	jammed	0.20	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	jammed	0.30	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	nominal	0.05	0.993 ± 0.014	0.993	0.000	0.000
uncertainty	nominal	0.10	1.000 ± 0.000	1.000	0.000	0.000
uncertainty	nominal	0.15	1.000 ± 0.000	1.000	0.000	0.000
uncertainty	nominal	0.20	1.000 ± 0.000	1.000	0.000	0.000
uncertainty	nominal	0.30	1.000 ± 0.000	1.000	0.000	0.000
uncertainty	ood-tail	0.05	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	ood-tail	0.10	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	ood-tail	0.15	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	ood-tail	0.20	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	ood-tail	0.30	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	rare-combo	0.05	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	rare-combo	0.10	0.000 ± 0.000	0.000	0.000	0.000
uncertainty	rare-combo	0.15	0.007 ± 0.014	0.007	0.000	0.000
uncertainty	rare-combo	0.20	0.021 ± 0.020	0.021	0.000	0.000
uncertainty	rare-combo	0.30	0.049 ± 0.025	0.056	0.000	0.007
uncertainty	rare-slip	0.05	0.250 ± 0.123	0.250	0.000	0.000
uncertainty	rare-slip	0.10	0.681 ± 0.154	0.694	0.007	0.014
uncertainty	rare-slip	0.15	0.861 ± 0.080	0.875	0.007	0.014
uncertainty	rare-slip	0.20	0.903 ± 0.064	0.931	0.007	0.028
uncertainty	rare-slip	0.30	0.944 ± 0.036	0.972	0.007	0.028

E STRESS APPENDIX

Table 12: All stress-sweep summary rows.

Method	Stress	Success	Macro F1	Rare recall	Tail	Safety
cal-risk	0.00	1.000 ± 0.000	0.000	0.000	0.000	0.000
cal-risk	0.20	0.950 ± 0.052	0.002	0.008	0.050	0.025
cal-risk	0.40	0.562 ± 0.143	0.068	0.014	0.438	0.275
cal-risk	0.60	0.062 ± 0.052	0.201	0.085	0.950	0.838
cal-risk	0.80	0.000 ± 0.000	0.455	0.547	1.000	0.775
cal-risk	1.00	0.013 ± 0.025	0.494	0.807	0.988	0.713
fail-active	0.00	1.000 ± 0.000	0.000	0.000	0.000	0.000
fail-active	0.20	0.950 ± 0.052	0.005	0.000	0.050	0.025
fail-active	0.40	0.562 ± 0.143	0.066	0.013	0.438	0.275
fail-active	0.60	0.062 ± 0.052	0.197	0.081	0.950	0.838
fail-active	0.80	0.000 ± 0.000	0.460	0.558	1.000	0.775
fail-active	1.00	0.013 ± 0.025	0.494	0.807	0.988	0.713
fs-v4	0.00	1.000 ± 0.000	0.000	0.000	0.000	0.000
fs-v4	0.20	0.950 ± 0.052	0.019	0.000	0.050	0.025
fs-v4	0.40	0.550 ± 0.143	0.086	0.002	0.450	0.300
fs-v4	0.60	0.075 ± 0.049	0.235	0.117	0.938	0.825
fs-v4	0.80	0.000 ± 0.000	0.493	0.655	1.000	0.850
fs-v4	1.00	0.013 ± 0.025	0.497	0.841	0.988	0.713
fs-v5	0.00	1.000 ± 0.000	0.008	0.000	0.000	0.000
fs-v5	0.20	0.950 ± 0.052	0.026	0.000	0.050	0.025
fs-v5	0.40	0.550 ± 0.143	0.107	0.005	0.450	0.300
fs-v5	0.60	0.075 ± 0.049	0.255	0.134	0.938	0.825
fs-v5	0.80	0.000 ± 0.000	0.471	0.606	1.000	0.850
fs-v5	1.00	0.013 ± 0.025	0.494	0.831	0.988	0.713
hybrid	0.00	1.000 ± 0.000	0.003	0.000	0.000	0.000
hybrid	0.20	0.950 ± 0.052	0.025	0.000	0.050	0.025
hybrid	0.40	0.562 ± 0.143	0.111	0.014	0.438	0.275
hybrid	0.60	0.062 ± 0.052	0.248	0.135	0.950	0.838
hybrid	0.80	0.000 ± 0.000	0.448	0.542	1.000	0.775
hybrid	1.00	0.013 ± 0.025	0.487	0.771	0.988	0.713
oracle-strata	0.00	1.000 ± 0.000	0.021	0.000	0.000	0.000
oracle-strata	0.20	0.950 ± 0.052	0.059	0.083	0.050	0.025
oracle-strata	0.40	0.562 ± 0.143	0.108	0.042	0.438	0.275
oracle-strata	0.60	0.062 ± 0.052	0.303	0.221	0.950	0.838
oracle-strata	0.80	0.000 ± 0.000	0.520	0.762	1.000	0.775
oracle-strata	1.00	0.013 ± 0.025	0.503	0.860	0.988	0.713
uncertainty	0.00	1.000 ± 0.000	0.003	0.000	0.000	0.000
uncertainty	0.20	0.950 ± 0.052	0.048	0.000	0.050	0.025
uncertainty	0.40	0.562 ± 0.143	0.108	0.009	0.438	0.275
uncertainty	0.60	0.062 ± 0.052	0.243	0.134	0.950	0.838
uncertainty	0.80	0.000 ± 0.000	0.475	0.606	1.000	0.775
uncertainty	1.00	0.013 ± 0.025	0.500	0.838	0.988	0.713

F PER-ROUND DATA-EFFICIENCY APPENDIX

Table 13: Combined-tail acquisition learning curves by round. These rows report data efficiency rather than final-only performance.

Method	Round	Success	Rare recall	Coverage
replay	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
replay	1	0.674 $\pm$ 0.048	0.613 $\pm$ 0.049	0.906 $\pm$ 0.040
replay	2	0.674 $\pm$ 0.048	0.612 $\pm$ 0.047	0.906 $\pm$ 0.040
replay	3	0.674 $\pm$ 0.048	0.615 $\pm$ 0.041	0.906 $\pm$ 0.040
replay	4	0.674 $\pm$ 0.048	0.615 $\pm$ 0.039	0.906 $\pm$ 0.040
replay	5	0.674 $\pm$ 0.048	0.614 $\pm$ 0.035	0.906 $\pm$ 0.040
cal-risk	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
cal-risk	1	0.674 $\pm$ 0.048	0.609 $\pm$ 0.048	0.906 $\pm$ 0.040
cal-risk	2	0.674 $\pm$ 0.048	0.606 $\pm$ 0.045	0.906 $\pm$ 0.040
cal-risk	3	0.674 $\pm$ 0.048	0.598 $\pm$ 0.045	0.906 $\pm$ 0.040
cal-risk	4	0.674 $\pm$ 0.048	0.597 $\pm$ 0.040	0.906 $\pm$ 0.040
cal-risk	5	0.674 $\pm$ 0.048	0.600 $\pm$ 0.034	0.906 $\pm$ 0.040
fail-active	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
fail-active	1	0.674 $\pm$ 0.048	0.605 $\pm$ 0.050	0.906 $\pm$ 0.040
fail-active	2	0.674 $\pm$ 0.048	0.599 $\pm$ 0.047	0.906 $\pm$ 0.040
fail-active	3	0.674 $\pm$ 0.048	0.596 $\pm$ 0.039	0.906 $\pm$ 0.040
fail-active	4	0.674 $\pm$ 0.048	0.601 $\pm$ 0.039	0.906 $\pm$ 0.040
fail-active	5	0.674 $\pm$ 0.048	0.602 $\pm$ 0.037	0.906 $\pm$ 0.040
fs-v4	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
fs-v4	1	0.681 $\pm$ 0.049	0.623 $\pm$ 0.037	0.906 $\pm$ 0.040
fs-v4	2	0.688 $\pm$ 0.054	0.631 $\pm$ 0.029	0.906 $\pm$ 0.040
fs-v4	3	0.688 $\pm$ 0.054	0.624 $\pm$ 0.034	0.906 $\pm$ 0.040
fs-v4	4	0.688 $\pm$ 0.054	0.618 $\pm$ 0.033	0.906 $\pm$ 0.040
fs-v4	5	0.688 $\pm$ 0.054	0.628 $\pm$ 0.029	0.922 $\pm$ 0.045
fs-v5	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
fs-v5	1	0.681 $\pm$ 0.049	0.611 $\pm$ 0.040	0.953 $\pm$ 0.045
fs-v5	2	0.681 $\pm$ 0.049	0.606 $\pm$ 0.040	0.953 $\pm$ 0.045
fs-v5	3	0.688 $\pm$ 0.054	0.604 $\pm$ 0.037	0.953 $\pm$ 0.045
fs-v5	4	0.674 $\pm$ 0.048	0.597 $\pm$ 0.034	0.969 $\pm$ 0.040
fs-v5	5	0.688 $\pm$ 0.054	0.604 $\pm$ 0.037	0.969 $\pm$ 0.040

Table 13 continued

Method	Round	Success	Rare recall	Coverage
hgb-active	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
hgb-active	1	0.674 $\pm$ 0.048	0.620 $\pm$ 0.036	0.906 $\pm$ 0.040
hgb-active	2	0.674 $\pm$ 0.048	0.618 $\pm$ 0.037	0.906 $\pm$ 0.040
hgb-active	3	0.674 $\pm$ 0.048	0.617 $\pm$ 0.035	0.906 $\pm$ 0.040
hgb-active	4	0.674 $\pm$ 0.048	0.612 $\pm$ 0.037	0.906 $\pm$ 0.040
hgb-active	5	0.674 $\pm$ 0.048	0.603 $\pm$ 0.039	0.906 $\pm$ 0.040
oracle-success	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
oracle-success	1	0.674 $\pm$ 0.048	0.621 $\pm$ 0.045	0.922 $\pm$ 0.045
oracle-success	2	0.674 $\pm$ 0.048	0.635 $\pm$ 0.044	0.938 $\pm$ 0.046
oracle-success	3	0.674 $\pm$ 0.048	0.652 $\pm$ 0.048	0.938 $\pm$ 0.046
oracle-success	4	0.674 $\pm$ 0.048	0.645 $\pm$ 0.040	0.953 $\pm$ 0.045
oracle-success	5	0.674 $\pm$ 0.048	0.647 $\pm$ 0.041	0.953 $\pm$ 0.045
hybrid	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
hybrid	1	0.681 $\pm$ 0.049	0.605 $\pm$ 0.045	0.906 $\pm$ 0.040
hybrid	2	0.674 $\pm$ 0.048	0.610 $\pm$ 0.050	0.922 $\pm$ 0.045
hybrid	3	0.674 $\pm$ 0.048	0.608 $\pm$ 0.051	0.922 $\pm$ 0.045
hybrid	4	0.674 $\pm$ 0.048	0.608 $\pm$ 0.051	0.922 $\pm$ 0.045
hybrid	5	0.674 $\pm$ 0.048	0.605 $\pm$ 0.041	0.922 $\pm$ 0.045
oracle-strata	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
oracle-strata	1	0.674 $\pm$ 0.048	0.724 $\pm$ 0.029	0.984 $\pm$ 0.031
oracle-strata	2	0.674 $\pm$ 0.048	0.690 $\pm$ 0.041	0.984 $\pm$ 0.031
oracle-strata	3	0.674 $\pm$ 0.048	0.656 $\pm$ 0.045	0.984 $\pm$ 0.031
oracle-strata	4	0.674 $\pm$ 0.048	0.642 $\pm$ 0.046	0.984 $\pm$ 0.031
oracle-strata	5	0.674 $\pm$ 0.048	0.637 $\pm$ 0.049	0.984 $\pm$ 0.031
rf-active	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
rf-active	1	0.674 $\pm$ 0.048	0.610 $\pm$ 0.048	0.906 $\pm$ 0.040
rf-active	2	0.674 $\pm$ 0.048	0.607 $\pm$ 0.048	0.906 $\pm$ 0.040
rf-active	3	0.674 $\pm$ 0.048	0.610 $\pm$ 0.047	0.906 $\pm$ 0.040
rf-active	4	0.674 $\pm$ 0.048	0.610 $\pm$ 0.047	0.906 $\pm$ 0.040
rf-active	5	0.674 $\pm$ 0.048	0.610 $\pm$ 0.045	0.906 $\pm$ 0.040

Table 13 continued

Method	Round	Success	Rare recall	Coverage
random	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
random	1	0.674 $\pm$ 0.048	0.612 $\pm$ 0.042	0.906 $\pm$ 0.040
random	2	0.674 $\pm$ 0.048	0.612 $\pm$ 0.044	0.906 $\pm$ 0.040
random	3	0.674 $\pm$ 0.048	0.622 $\pm$ 0.039	0.906 $\pm$ 0.040
random	4	0.674 $\pm$ 0.048	0.622 $\pm$ 0.042	0.906 $\pm$ 0.040
random	5	0.674 $\pm$ 0.048	0.631 $\pm$ 0.046	0.906 $\pm$ 0.040
coreset	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
coreset	1	0.674 $\pm$ 0.048	0.609 $\pm$ 0.041	0.984 $\pm$ 0.031
coreset	2	0.674 $\pm$ 0.048	0.614 $\pm$ 0.040	0.984 $\pm$ 0.031
coreset	3	0.674 $\pm$ 0.048	0.621 $\pm$ 0.040	0.984 $\pm$ 0.031
coreset	4	0.674 $\pm$ 0.048	0.623 $\pm$ 0.042	0.984 $\pm$ 0.031
coreset	5	0.674 $\pm$ 0.048	0.618 $\pm$ 0.041	0.984 $\pm$ 0.031
tail-risk	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
tail-risk	1	0.674 $\pm$ 0.048	0.606 $\pm$ 0.049	0.906 $\pm$ 0.040
tail-risk	2	0.674 $\pm$ 0.048	0.601 $\pm$ 0.046	0.906 $\pm$ 0.040
tail-risk	3	0.674 $\pm$ 0.048	0.609 $\pm$ 0.041	0.906 $\pm$ 0.040
tail-risk	4	0.674 $\pm$ 0.048	0.606 $\pm$ 0.042	0.906 $\pm$ 0.040
tail-risk	5	0.674 $\pm$ 0.048	0.600 $\pm$ 0.044	0.906 $\pm$ 0.040
task-label	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
task-label	1	0.681 $\pm$ 0.049	0.611 $\pm$ 0.049	0.906 $\pm$ 0.040
task-label	2	0.681 $\pm$ 0.049	0.618 $\pm$ 0.046	0.906 $\pm$ 0.040
task-label	3	0.681 $\pm$ 0.049	0.624 $\pm$ 0.044	0.906 $\pm$ 0.040
task-label	4	0.681 $\pm$ 0.049	0.632 $\pm$ 0.038	0.906 $\pm$ 0.040
task-label	5	0.681 $\pm$ 0.049	0.630 $\pm$ 0.039	0.906 $\pm$ 0.040
uncertainty	0	0.674 $\pm$ 0.048	0.613 $\pm$ 0.050	0.906 $\pm$ 0.040
uncertainty	1	0.674 $\pm$ 0.048	0.612 $\pm$ 0.048	0.906 $\pm$ 0.040
uncertainty	2	0.674 $\pm$ 0.048	0.617 $\pm$ 0.042	0.906 $\pm$ 0.040
uncertainty	3	0.674 $\pm$ 0.048	0.620 $\pm$ 0.046	0.906 $\pm$ 0.040
uncertainty	4	0.674 $\pm$ 0.048	0.621 $\pm$ 0.042	0.906 $\pm$ 0.040
uncertainty	5	0.674 $\pm$ 0.048	0.618 $\pm$ 0.037	0.906 $\pm$ 0.040

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